BADM 534

Data Analytics for Marketing Applications

Lecture Notes

Marketing Decisions and AI-Assisted Coding

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Chapter 1

Marketing Decisions, Data, and AI-Assisted Python

A retailer must decide which customers should receive a coupon, how much inventory to place in each store, and which product descriptions should appear online. Each decision needs a different analytical object. A forecast of next week's purchases can inform inventory. A comparison of purchases with and without a coupon can inform targeting. A summary of product complaints can identify a service problem. An analysis becomes useful when its output corresponds to the decision the manager can actually change.

1.1 From a Marketing Question to an Analytical Object

Consider a subscription service deciding whether to offer a renewal discount. The service observes account tenure, past payments, recent use, and prior contacts. A descriptive analysis reports renewal rates by tenure. A predictive analysis estimates the probability that an account renews under the conditions represented in the training data. A causal analysis estimates how an offer changes renewal. A decision analysis combines that change with margins, contact costs, and operational constraints. These objects can use overlapping data while answering different questions.

The timing of the decision determines what information is available. If an account must be selected on Monday morning, a support call made on Wednesday cannot be a predictor for that selection. A variable can be strongly associated with an outcome and still be unusable at the decision time. Write down a forecast origin, an outcome window, and a feature window before constructing a table. An account-week table and an account-lifetime table support different forecasts because their rows and time horizons differ.

For a renewal offer, a useful decision quantity is incremental expected contribution:

$$\Delta\pi_i = m_i\{p_i(1) - p_i(0)\} - c_i.$$

Here i identifies an account, $p_i(1)$ and $p_i(0)$ are its renewal probabilities with and without the offer, m_i is contribution in dollars per retained account over the specified horizon, and c_i is the expected incremental cost of making the offer. This compact expression assumes that the offer's costs can be summarized by c_i and that m_i already reflects relevant variable costs. When every renewing recipient receives a discount, discount expense depends on renewal under treatment and should be modeled explicitly.

The difference between the two renewal probabilities matters because many accounts would renew without assistance. Suppose contribution is \$100 and contact cost is \$3. Then an account whose renewal probability rises from 0.90 to 0.91 has incremental expected contribution of \$-2. An account whose probability rises from 0.40 to 0.50 has contribution of \$7. Ranking by treated

renewal probability chooses the first account; ranking by incremental contribution chooses the second. Observing only treated customers would not reveal the untreated probabilities.

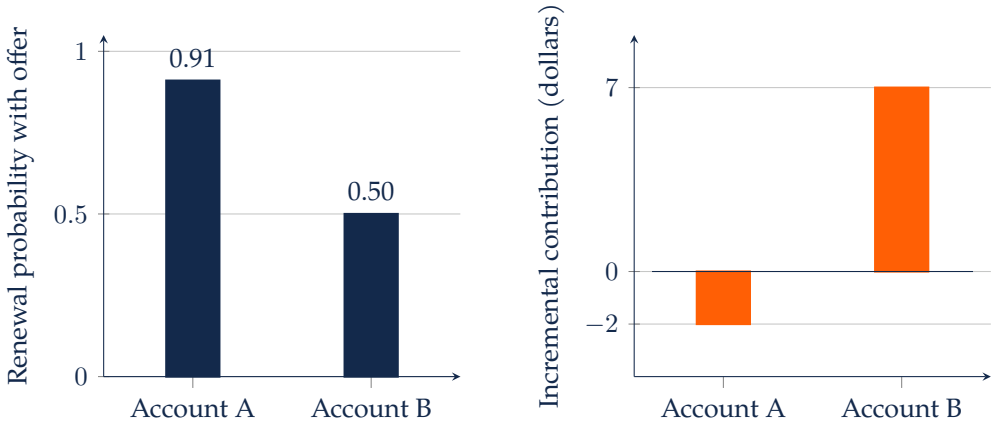


Figure 1.1. High response and high incremental value can identify different customers.

With contribution and contact cost held fixed, the first account's high baseline renewal probability reverses its ranking under incremental value. A response model and a targeting policy therefore need separate evaluation.

1.2 The Unit of Observation Determines the Calculation

A transaction system often stores one row per product line, while the marketing question concerns orders or customers. Suppose order O1 contains two product lines, O2 contains one, and O3 contains three. There are six rows, three orders, and perhaps only two customers. Counting rows as purchases would report six orders. Averaging customer attributes over product lines would give more weight to customers with larger baskets.

A key uniquely identifies the intended unit. At the line level, the key might be the combination of order ID and line number. At the order level, it is order ID. At the customer level, it is customer ID. A useful data contract states the key, allowed duplicates, measurement units, missing-value codes, and date conventions. It also states whether a negative quantity represents a return, whether a price includes tax, and whether canceled orders belong in the analysis.

For line ℓ in order o, gross line sales are quantity times unit price, $s_{o\ell} = q_{o\ell}p_{o\ell}$. Order sales are $S_o = \sum_\ell s_{o\ell}$, and customer-period sales are the sum across that customer's eligible orders during the period. These are identities once the population and variables have been defined. They do not require a behavioral assumption. The substantive choices enter through eligibility, returns, discounts, and the observation window.

Joining data can change the population or its weights. A many-to-one join from order lines to a customer table should add customer attributes without multiplying the original rows. If the customer table accidentally contains two records for a customer, every matching order line appears twice. A program can finish without an error while overstating sales. Verify the relationship between keys before joining, and compare row counts and totals afterward.

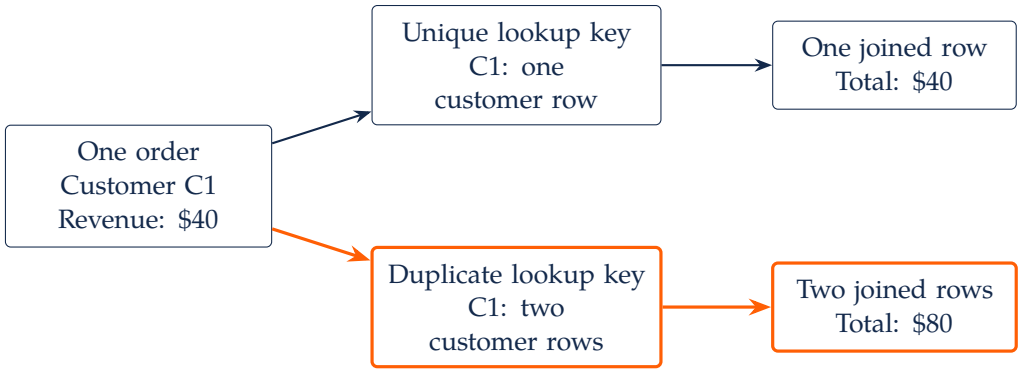


Figure 1.2. A duplicate customer key multiplies transactions in a join.

For a single $40 order, a customer lookup with two matching records produces two joined rows and an apparent $80 total. This error arises from the structure of the tables. Increasing the sophistication of the subsequent model cannot repair it.

1.3 Python Objects and Reproducible Calculations

Python stores individual values in objects such as integers, floating-point numbers, strings, and booleans. A list stores an ordered collection; a dictionary associates names with values. A function packages a calculation with explicit inputs and an output. These distinctions matter when an assistant supplies code: a column of numeric-looking strings does not behave like a numeric column, and an empty string does not automatically represent a missing value.

Suppose two orders contain three product lines. Multiplying quantity by unit price gives line sales; summing within each order gives order sales. Returning each calculation as a number allows its result to be checked and reused.

```
lines = [
    {"order": "O1", "quantity": 2, "unit_price": 10.0},
    {"order": "O1", "quantity": 1, "unit_price": 5.0},
    {"order": "O2", "quantity": 3, "unit_price": 4.0},
]

def line_sales(line):
    return line["quantity"] * line["unit_price"]

order_sales = {}
for line in lines:
    key = line["order"]
    order_sales[key] = order_sales.get(key, 0.0) + line_sales(line)

assert order_sales == {"O1": 25.0, "O2": 12.0}
assert sum(order_sales.values()) == 37.0
```

The total is $37 across two orders, so average order value is $18.50. Dividing by the three product lines would instead give average line sales, $12.33. Neither denominator is inherently wrong; they describe different objects. The mistake would be assigning one calculation the other's interpretation.

A DataFrame provides a tabular interface for larger data. In pandas, selecting columns, filtering rows, creating variables, grouping observations, and joining tables are distinct operations. Use explicit intermediate names when those operations change the unit of observation. For example, `eligible_lines`, `orders`, and `customer_months` communicate more than repeatedly overwriting `df`. The pandas tutorials provide syntax references; the analytical specification should determine which operations are needed.

Notebook execution order can hide dependencies. A cell may work because an earlier interactive command created a variable that is absent from the saved notebook. Before describing a notebook as reproducible, restart the kernel and run all cells in order. Store file paths relative to the project, record package versions, and distinguish source data from derived outputs. A random seed helps reproduce a simulation or split, but it does not guarantee identical results across every software version or computing platform.

1.4 Case Study: A Coupon Campaign Specification

A retailer has a monthly contact budget for 5,000 customers. The proposed campaign offers a discount on the next purchase. The manager initially asks for "a model to find our best customers." That request leaves the action, outcome, and objective unresolved. The highest-spending customers may receive the offer without changing their behavior; customers likely to respond may buy products with low margins.

A usable specification would identify the decision date, eligible customer population, offer, evaluation horizon, and objective. One possibility is to maximize incremental contribution during the next 30 days, subject to the contact limit. Eligibility might require marketing consent and an active

account. The analysis should define contribution after discounts, returns, fulfillment, and contact costs. It should identify which customer information existed before assignment and how untreated outcomes will be measured.

The initial dataset might include customer-level histories and a randomized pilot. Without such a pilot or another credible design, a response model can help describe or predict behavior, but the incremental contribution calculation remains unresolved. The next operational step may therefore be a randomized test, rather than immediate large-scale optimization.

A specification also needs acceptance criteria. The eligible-customer table must have one row per customer, no post-assignment features, and the intended date coverage. A campaign summary must reconcile assigned counts, contact costs, purchases, and contribution. These checks are small enough to state before asking an AI assistant to implement the analysis. They protect the meaning of the result while allowing flexibility in the code.

1.5 AI-Assisted Coding as a Sequence of Verifiable Steps

An effective coding request names the inputs, output, definitions, and checks. For the transaction example, ask for a function that accepts product-line records, validates required fields, aggregates sales by order, and returns both order totals and an exception report. Provide the three-line example and its expected $37 total. Ask the assistant to explain what happens with a return, a missing order ID, or a duplicated line.

Generate one analytical transformation at a time when the next step depends on a definition. Inspect the proposed code before executing it, especially commands that install software, delete files, upload data, or invoke an external API. Read error messages as evidence about the code's assumptions. A type error may reveal numeric values stored as strings; a merge error may reveal an incorrect key. Repairing the error without understanding its cause can preserve the underlying analytical mistake.

Verification should use both identities and meaningful counterexamples. Order totals must sum to the line total for the same eligible rows. A single known customer should produce a manually calculable summary. An empty input should have an explicit result. A missing customer lookup should not silently remove sales. These checks constrain what the implementation is allowed to do without forcing one particular coding style.

Keep a short record of consequential AI contributions: the task requested, important changes to the plan or code, and the checks performed. A saved conversation alone does not establish correctness. The record should make it possible to explain why a transformation was accepted.

1.6 Interpreting and Communicating a Result

A useful recommendation names the decision, states the relevant estimate with units and uncertainty, and explains how it changes the action. "The model has high accuracy" leaves the manager's decision unresolved. "At a $3 contact cost and the stated contribution assumptions, contact the customers with positive estimated incremental contribution, subject to capacity" connects the

analysis to the objective. The recommendation still needs evidence about how the probabilities were estimated and whether the policy was evaluated on new customers or periods.

Results should distinguish observation from assumption. A transaction total is calculated from recorded eligible rows. A renewal probability is an estimate. A contribution margin may be supplied by finance. An untreated outcome for an offered customer is counterfactual. Listing these roles prevents a spreadsheet or notebook from giving every input the appearance of equal certainty.

If one customer places four orders and another places one, an order-weighted mean gives the first customer four times the weight. If both customers should count equally, calculate each customer's mean first. If an AI-generated program reports six purchases from six product lines, inspect whether purchase was defined as a line or an order. If an offer has a high response rate, ask for the response rate under no offer before interpreting the response as an effect.

1.7 Readings

The Python marketing research companion repository provides examples tied to marketing applications. Its notebooks are reference material and require inspection of dependencies and data paths before adaptation. The pandas tutorials support tabular implementation.

Chapter 2

Customer Data, Measurement, and Visualization

A grocery retailer sees larger online baskets and asks whether moving customers to its app will raise sales. The raw comparison combines the shopping channel with who uses it, why they shop, and which trips are observed. A household may order online for a planned weekly restock and visit a store for a forgotten ingredient. Measurement and visualization determine whether the analysis reveals that distinction or conceals it.

2.1 Define the Population Before the Metric

A customer dataset is a record of an observation process. Loyalty accounts exclude unidentified shoppers. Online logs record sessions that may contain repeated visits by the same person. Orders exclude browsing without purchase. App events reflect tracking rules, consent, device settings, and instrumentation changes. Before calculating a metric, identify the target population and the mechanism by which its members enter the table.

Suppose monthly revenue per customer is intended to describe all eligible accounts. Dividing total revenue by purchasing accounts produces revenue per purchasing account. If inactivity changes, that measure can increase while revenue per eligible account falls. The denominator should therefore be written into the metric name or its definition. For customer acquisition cost, specify whether the denominator is new registered accounts, first-time purchasers, or customers with positive contribution over a defined horizon.

A conversion rate requires an eligible exposure count. If b eligible sessions produce at least one purchase among n eligible sessions, session conversion is b/n. Customer conversion instead uses customers in both numerator and denominator. A customer with ten sessions should not receive ten times the weight in a customer-level rate. When the same customer contributes repeated observations, statistical uncertainty must also account for that dependence.

Returns require a substantive rule. Gross sales, net sales after returns, and contribution after variable costs answer different questions. A comparison that deducts online returns but observes only gross offline sales mixes measurement rules. Currency, tax, fulfillment fees, and discounts should be aligned across channels before interpreting differences.

2.2 Build a Customer Table Without Losing Customers

Begin with a customer roster that defines eligibility. Aggregate eligible transactions to customer-period totals, then left-join those totals to the roster. A customer with no transactions can receive zero sales if the transaction system fully captures activity during the observation period. Zero should not replace a value that is missing because the customer was unobserved or the data feed failed.

A join should have a documented relationship: one-to-one, many-to-one, or one-to-many. Validate key uniqueness in the lookup table and inspect unmatched records. A match rate of 99% can still conceal a material problem if the unmatched 1% contains the highest-value customers or one entire channel. Report both counts and relevant value totals for unmatched observations.

Dates need similar attention. Convert timestamps to a consistent timezone before assigning day or week. Determine whether the date represents ordering, shipment, payment, or return. A promotion announced late in a period may affect orders immediately but recognized revenue later. Aligning metrics on an inappropriate date can create an apparent lag that comes from accounting rather than customer behavior.

Outliers should be investigated before removal. A $10,000 order could be a business customer, a data-entry error, or a legitimate event. A rule that trims all orders above a percentile changes the estimand to a restricted population. State that restriction and compare conclusions with and without the rule when it can affect the decision. Robust summaries such as medians and quantiles often reveal more than one mean with an undocumented exclusion.

2.3 Case Study: A Reversal Caused by Customer Mix

A marketing manager compares campaigns A and B using conversion. Both campaigns reach new and returning customers, but their audience composition differs. Assume the following customer counts and outcomes, with each customer belonging to one segment and appearing once.

Campaign	New Customers	New Conversions	Returning Customers	Returning Conversions	Overall Conversion
A	100	9	100	80	89/200 = 44.5%
B	200	20	100	81	101/300 = 33.7%

Campaign B has higher conversion among new customers, 10% versus 9%, and among returning customers, 81% versus 80%. Campaign A nevertheless has the higher aggregate rate because half its audience consists of returning customers, compared with one-third for B. Returning customers convert much more frequently in both campaigns.

The aggregate rate is a weighted average of segment rates:

$$\bar{p}_j = \sum_g w_{jg} p_{jg}, \qquad \sum_g w_{jg} = 1.$$

Here j indexes campaigns, g indexes customer segments, p_{jg} is the conversion fraction within a segment, and w_{jg} is that segment's share of the campaign audience. The weights are part of the comparison. Holding them at 50% new and 50% returning gives standardized rates of 44.5% for A and 45.5% for B. This adjustment describes performance for the chosen common mix. It establishes a causal campaign effect only if the design supports that interpretation.

B's small advantages within both segments coexist with A's aggregate advantage. A manager deciding where to send the next campaign needs to specify the future audience mix and whether customers were assigned comparably. An aggregate bar chart alone cannot answer that question.

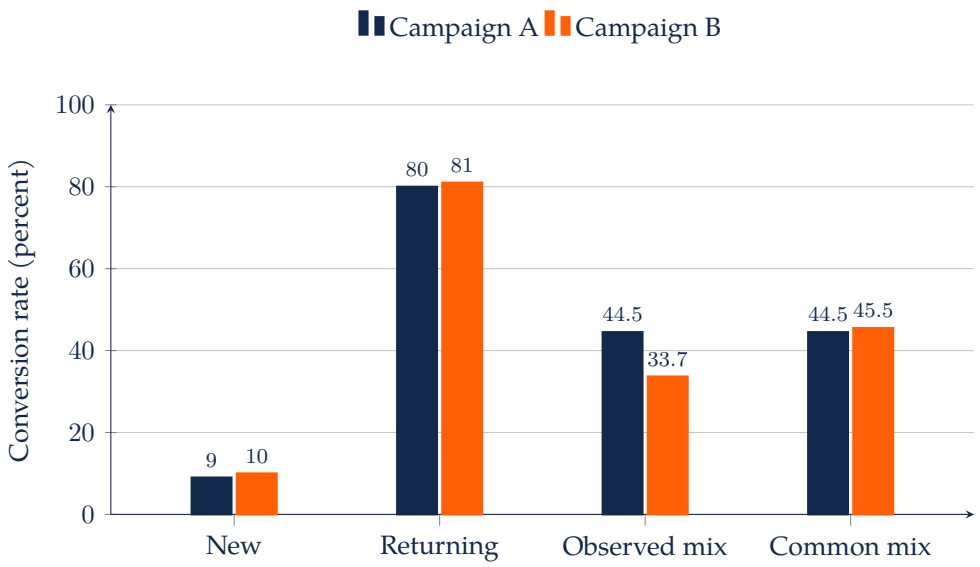


Figure 2.1. Audience composition reverses the aggregate campaign ranking.

2.4 Cohorts, Retention, and Customer Age

A cohort groups customers by a common starting event, such as first purchase or subscription activation. Retention at customer age k is the fraction of the original eligible cohort meeting a stated activity criterion at that age. A cohort table separates calendar time from customer age: an October signup observed one month later is not comparable with an April signup observed seven months later unless the question explicitly concerns those different ages.

For cohort c, define N_c as the number of eligible customers at entry and $A_{c,k}$ as the number active at age k. The descriptive retention rate is $R_{c,k} = A_{c,k}/N_c$. A later cohort may not yet have reached age k; that cell is unobserved, not zero. Omitting inactive customers from the denominator would turn retention into a different conditional measure.

Signup Cohort	Entry Customers	Active at Month 1	Active at Month 2	Active at Month 3
January	100	60	45	40
February	100	70	50	Not yet observed
March	100	80	Not yet observed	Not yet observed

Month-one retention rises from 60% for the January cohort to 80% for the March cohort. March customers have not yet reached the later observation windows, so their long-run retention remains unobserved. Comparing March's month one with January's month three would combine cohort quality with customer age. The right comparison depends on whether the decision concerns onboarding, long-term value, or a calendar-time change affecting all existing customers.

Unobserved future periods require forecasts from an explicit model. Coding those periods as zero would treat unavailable outcomes as observed failures and bias retention downward.

	At Entry	Month 1	Month 2	Month 3
January	100%	60%	45%	40%
February	100%	70%	50%	Not yet observed
March	100%	80%	Not yet observed	Not yet observed

Each cohort begins with 100 customers.
Only observed customer ages enter the retention denominator.

Figure 2.2. A cohort's unobserved future is different from zero retention.

2.5 Visual Encodings Should Match the Question

Use position on a common scale when precise comparison matters. A dot plot can compare segment conversion rates more clearly than area-sized bubbles. A line chart is appropriate when the ordering of time is meaningful. A histogram or empirical cumulative distribution reveals skewness that a mean hides. A scatterplot supports examination of a relationship, with transparency or aggregation when observations overlap.

The scale changes the interpretation. A logarithmic sales axis emphasizes proportional differences; a linear axis emphasizes dollar differences. A truncated bar-chart axis exaggerates differences because bar length encodes magnitude from its baseline. A carefully labeled narrow axis can be useful for a line or dot plot when the scientific question concerns small differences. The choice should preserve the intended comparison and disclose the scale.

Uncertainty should be shown for an estimated comparison when it affects a decision. For an independent Bernoulli sample, a common large-sample standard error for a proportion is $\sqrt{\hat{p}(1-\hat{p})/n}$. Near zero or one, or with small samples, use an interval appropriate to binomial data. With repeated observations per customer, treating every session as independent generally understates uncertainty. Resample or cluster at the unit that captures the relevant dependence.

Color can encode categories or highlight the comparison that matters. Use line styles, markers, and labels alongside color, and keep small text dark enough for contrast. A figure should remain interpretable when printed in grayscale or described to a student who cannot see it. Explain the axes, units, sample, and principal comparison in its caption or accompanying text.

2.6 Case Study: Online Grocery Baskets and Repeated Purchases

Chintala, Liaukonytė, and Yang's grocery-shopping study compares online and offline baskets using household shopping records. It studies basket composition and repeated purchasing, with attention to differences in shopping-trip types. Larger baskets need not contain a wider range of categories, and a household's choice of channel can be related to the purpose of its trip.

Suppose one household buys the sets {milk, bread, apples, yogurt} and {milk, bread, apples, coffee} on successive trips. Each basket contains four distinct items. Their Jaccard similarity is

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} = \frac{3}{5} = 0.60.$$

The numerator counts shared distinct items and the denominator counts all distinct items appearing in either trip. A different pair of four-item baskets with one shared item has similarity $1/7 \approx 0.143$. Basket size is identical in the two examples, while repeated purchasing differs substantially. Whether quantities or spending weights should enter the measure depends on the decision; set-based Jaccard treats one unit and ten units of an item alike.

A defensible descriptive study would define eligible grocery trips, compare channels within households when appropriate, and examine basket size, distinct categories, repeat-item overlap, and spending. It would inspect whether findings are driven by large restocking trips, small convenience trips, or particular retailers. Channel labels should not silently determine the definition of a "typical" trip when the objective is a comparable classification. Clustering can classify trips using their observed composition.

A retail recommendation might distinguish discovery from replenishment. High repetition can support convenient reordering while making it harder for unfamiliar products to receive attention. A proposed interface change should therefore specify both the desired outcome and the possible tradeoff. Its effect requires evaluation under an appropriate design, even when the descriptive pattern provides a strong reason to test it.

2.7 From Evidence to a Management Presentation

The Pyramid Principle organizes a recommendation around a decision-relevant answer, followed by a small number of supporting reasons and the evidence for each reason. This structure works when the answer's strength matches the analysis. A descriptive study might recommend a test or a closer diagnostic comparison, rather than immediate deployment.

For the campaign example, a clear opening is: "Compare the campaigns at a common customer mix before reallocating the budget." The supporting reasons are that returning customers convert more often, the campaigns reached different mixes, and the within-segment ranking differs from the aggregate ranking. The supporting evidence should quantify each reason and connect it to the allocation decision.

An effective chart title can state the comparison, such as "Campaign A's aggregate advantage reflects its returning-customer share." The caption should define the conversion denominator and identify the data as synthetic.

2.8 AI-Assisted Workflow and Practice

Ask an assistant to construct a customer-period table from a specified roster and transaction table, preserving customers with no observed purchases. Require a join audit, a missing-data report, and a reconciliation of total sales. Next ask it to calculate the campaign table above and standardized rates under a 50/50 audience mix. Verify those small numbers before applying the same functions to a larger file.

For a chart request, specify the question, observation unit, axis units, ordering, and uncertainty representation. Ask the assistant to describe what visual pattern would change the recommendation. Compare its description with the plotted data. If it labels an observational channel difference as an effect, revise the interpretation and identify the design that would be needed to support the stronger claim.

For practice, change campaign B's audience to 100 new and 100 returning customers while keeping its segment conversion rates fixed. Its aggregate rate becomes 45.5%, and the reversal disappears. Next calculate month-two retention for the January and February cohorts: 45% and 50%. March has no observed month-two rate. Finally, compare a four-item basket with itself: Jaccard similarity is one, regardless of whether the quantities have changed. Each exercise tests a definition that should remain explicit in an automated analysis.

2.9 Readings

Wilke's visualization text offers a visual reference for distributions, comparisons, and uncertainty. The pandas grouping documentation supports implementation.

Chapter 3

Regression, Interpretation, and Marketing Decisions

A retailer observes that stores with larger advertising budgets sell more. The manager wants to know whether another dollar of advertising would be profitable. The observed relationship combines the response to advertising with differences in store size, local demand, seasonality, and the firm's reasons for choosing the budget. Regression provides a way to summarize conditional relationships. Its value depends on which variation identifies the comparison and whether that comparison answers the manager's question.

3.1 A Conditional Mean and a Regression Line

Let Y_i be weekly store sales measured in thousands of dollars and X_i be weekly advertising expenditure in thousands of dollars. A simple linear model writes

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i.$$

The intercept β_0 is the fitted sales level at zero advertising; its practical interpretation is weak when zero lies far outside the observed range. The slope β_1 is the change in fitted sales, in thousands of dollars, associated with an additional thousand dollars of advertising. The residual captures the difference between observed sales and the fitted line. Linearity is an approximation to the relationship over the relevant range, not a claim that every customer's behavior is linear.

Ordinary least squares chooses coefficients to minimize the sum of squared residuals. In a simple regression with an intercept, its slope is

$$\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2}.$$

The numerator measures whether high-advertising observations tend to have high sales. The denominator measures how much advertising varies. When advertising hardly varies, the data provide little information about a slope, even if the sample contains many rows. A large sample of nearly identical budgets does not substitute for informative variation.

For three observations $(X, Y) = (1, 3), (2, 5), (3, 7)$, the fitted line is $\hat{Y} = 1 + 2X$. At $X = 2.5$, it predicts sales of \$6,000. If the sales contribution margin is 30%, an additional \$1,000 of advertising is associated with \$600 of contribution from \$2,000 of extra sales. Treating that association as a return to intervention requires an identification argument. Even under a causal interpretation, the incremental contribution is less than the additional advertising expense in this numerical example.

3.2 Multiple Regression and the Comparison It Uses

A multiple regression can include observed store characteristics:

$$Y_i = \beta_0 + \beta_1 X_i + \gamma Z_i + \varepsilon_i,$$

where Z_i might be floor area measured in thousands of square feet. The advertising coefficient summarizes variation in sales associated with advertising after removing the linear association with floor area. A useful way to understand this is to regress advertising on the other controls, regress sales on the same controls, and regress the resulting sales residuals on the advertising residuals. The Frisch-Waugh-Lovell result gives the same advertising coefficient as the full regression when the residualization includes the same intercept and controls.

"Controlling for" a variable therefore changes the comparison. It does not automatically remove all confounding. If managers observe local demand information that the analyst lacks, their advertising choices may remain related to unobserved sales determinants. Controls measured after the intervention can also remove part of the effect or introduce bias. Select variables using the decision timing and a substantive account of how the data were generated.

When an omitted variable affects both the predictor and the outcome, the simple slope can be misleading. In the linear population model above, suppose the regression disturbance is mean-independent of X and Z. Omitting Z yields a simple-regression slope equal to

$$\beta_1 + \gamma \frac{\text{Cov}(X, Z)}{\text{Var}(X)}.$$

The extra term has a substantive interpretation. If larger stores sell more, so $\gamma > 0$, and larger stores spend more on advertising, so the covariance is positive, the simple advertising slope exceeds the conditional slope. The equation is a model-based decomposition; its interpretation depends on the stated linear specification and exogeneity condition.

3.3 Case Study: Attractive Markets and Apparent Competitive Benefits

Igami and Yang (2016) study entry and competition among hamburger chains in Canada. Their setting motivates a general identification problem: attractive markets can support both more competitors and better outcomes for a focal firm. A positive relationship between rival presence and performance can therefore coexist with a negative competitive effect.

Within low-demand markets, suppose stores facing one, two, and three rivals have sales of 9, 8, and 7 units. Within high-demand markets, stores facing four, five, and six rivals have sales of 16, 15, and 14 units. Within each demand group, one additional rival is associated with one unit less sales. Across the six observations, the pooled slope is approximately $+1.57$: high-demand markets have more rivals and higher sales.

The within-group slope is -1, while baseline demand differs between groups. The positive pooled slope reflects that demand composition. In an actual retail application, market demand may be unobserved, entry may be forward-looking, and firms may respond strategically.

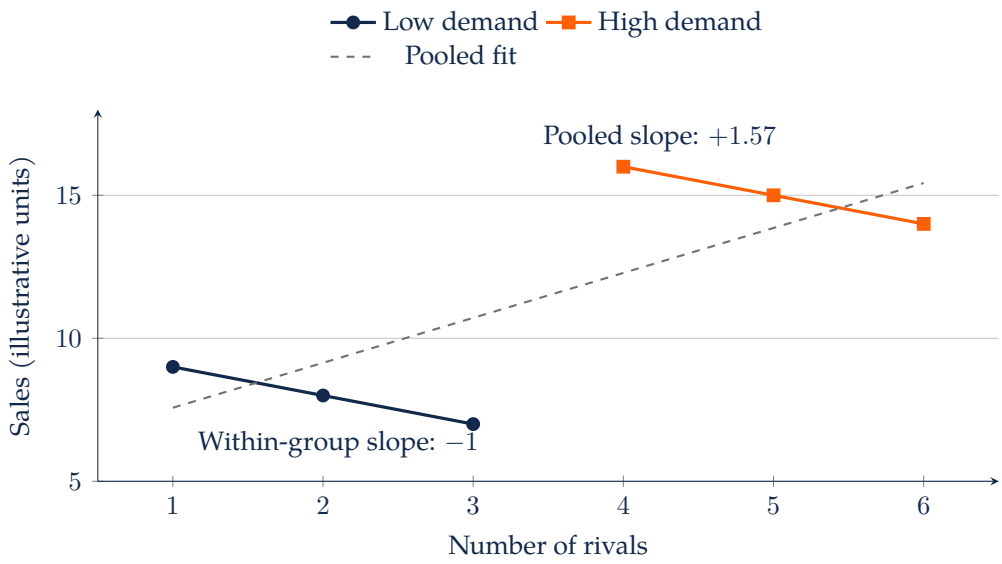


Figure 3.1. A positive pooled relationship can conceal a negative within-market-type relationship.

A manager considering entry should ask which outcomes would change if an additional competitor entered the same market. Comparing different markets does not automatically provide that counterfactual. Market fixed effects, measured demand controls, an instrument, or a credible event design may improve the comparison, but each uses different variation and requires its own conditions. A highly predictive regression can remain unsuitable for interpreting the coefficient as competition's causal effect.

3.4 Transformations and Economically Meaningful Units

A level-level specification gives an additive interpretation: one more unit of X changes fitted Y by β_1. A log-level specification, $\log Y = \beta_0 + \beta_1 X + \varepsilon$, describes proportional differences in a positive outcome. A one-unit change in X changes the fitted log outcome by β_1; the corresponding multiplicative change in the log-scale prediction is e^{β_1}, or $100(e^{\beta_1} - 1)$ percent. The approximation $100\beta_1$ percent is useful when the coefficient is small.

A log-log demand model, $\log Q = \alpha + \beta \log P + \varepsilon$, gives β an elasticity interpretation over the range where the specification applies. If $\beta = -1.5$, a 1% higher price is associated with approximately 1.5% lower quantity. Sales revenue is PQ, so its local price elasticity is $1 + \beta = -0.5$. Profit also depends on unit cost. A price that raises revenue can still lower profit, and observed prices can be endogenous because firms set them in response to demand.

Zero and negative outcomes need deliberate treatment. Taking a logarithm of zero is undefined. Adding one changes the scale and interpretation, especially when values are small or the unit of measurement changes. A two-part model can separate purchase incidence from positive expenditure. A suitable generalized linear model may model the conditional mean directly. The choice should reflect the outcome and intended estimand, with diagnostics for the selected specification.

Back-transforming a predicted log outcome also needs care. Exponentiating the predicted conditional mean of $\log Y$ generally does not produce the conditional mean of Y, because exponenti-

ation is nonlinear. Retransformation methods or a model for the mean on the original scale may be needed when total sales is the target. Evaluate predictions in the units used for the managerial decision.

3.5 Interactions Describe How a Relationship Changes

Suppose an analyst expects a promotional message to work differently for customers with long tenure. A linear interaction specification is

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 T_i + \beta_3 D_i T_i + \varepsilon_i,$$

where D_i is a message indicator and T_i is tenure in years. The fitted message difference at tenure t is $\beta_1 + \beta_3 t$. The coefficient β_1 is the difference at zero tenure, so centering tenure at a meaningful value can make that coefficient easier to interpret. The interaction coefficient tells how the fitted difference changes per year of tenure.

With $\beta_1 = 4$ dollars and $\beta_3 = -1$ dollar per year, the fitted difference is \$4 at tenure zero, \$2 at two years, and zero at four years. An average coefficient can conceal that variation. Predictions beyond the observed tenure range should be treated as extrapolation. In a nonlinear probability model, an interaction coefficient on the model's index does not directly equal the interaction in probability; compute and plot the predicted probabilities or marginal contrasts.

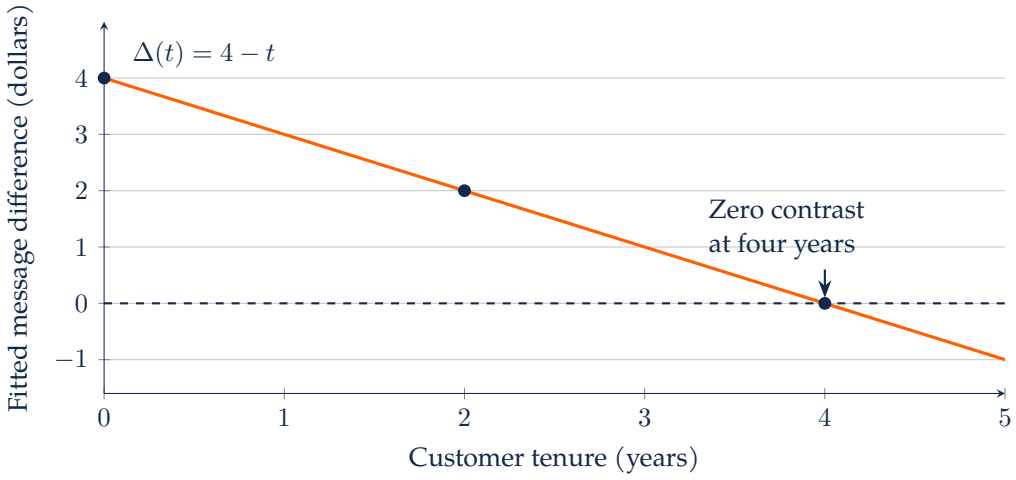


Figure 3.2. An interaction changes the relevant comparison across customer tenure.

For tenure between zero and five years, the assumed contrast $4 - t$ crosses zero at four years. In an estimated application, uncertainty around the contrast and support at each tenure would determine whether such a crossing is informative.

3.6 Uncertainty, Dependence, and Practical Importance

A standard error describes sampling uncertainty under an estimator and its assumptions. A confidence interval provides a range generated by a procedure with a stated repeated-sampling cov-

erage property. A p-value assesses compatibility with a particular null under its reference model. None of these quantities measures the managerial value of an effect by itself. A precisely estimated increase of one cent may be unimportant; a wide interval around a large potential benefit may justify another experiment.

For an approximately normal coefficient estimate, a rough 95% interval is the estimate plus or minus 1.96 standard errors, with appropriate adjustments when a finite-sample reference distribution is needed. If a sales slope is 2.0 with a standard error of 0.5, that interval is approximately $[1.02, 2.98]$ in the slope's units. Report the units and the comparison alongside the interval.

Heteroskedasticity means that residual variance differs across observations. Heteroskedasticity-robust standard errors can address that feature for inference under suitable sampling conditions; they do not repair omitted-variable bias. Repeated store-week observations may share persistent store shocks. Clustered standard errors allow dependence within a specified cluster under their asymptotic conditions. The relevant number of independent clusters can be much smaller than the number of rows. With few clusters, conventional approximations may be unreliable and design-specific methods are needed.

3.7 Diagnostics and Panel Data

Inspect residuals against fitted values and important predictors for nonlinear patterns and changing variance. Investigate influential observations, errors in units, and unusual leverage. Multicollinearity can make individual coefficients unstable even when predictions are reasonably stable, because several predictors explain nearly the same variation. Removing a variable solely to obtain a smaller p-value changes the question and can introduce omitted-variable bias.

Store fixed effects absorb time-invariant differences between stores. Calendar fixed effects absorb common period-specific shocks. In a store-week panel, a model with both uses changes within stores relative to common time movements. It still requires a valid account of time-varying confounders. If a store raises advertising precisely when an unobserved local event increases demand, the advertising coefficient can remain biased.

A fitted model should be evaluated on the task for which it is used. For prediction, compare held-out errors with a useful baseline. For a descriptive coefficient, inspect support and sensitivity to substantively justified specifications. For a causal effect, assess the design and identifying conditions. A large R^2 indicates in-sample fit relative to an intercept-only benchmark; it does not establish a credible intervention comparison.

3.8 AI-Assisted Implementation and Practice

Ask an assistant to fit the pooled and demand-group-adjusted regressions in the six-observation example. Require a table with variable definitions, units, coefficients, and the exact sample. Verify that the adjusted rival coefficient is -1 and the pooled coefficient is $11/7 \approx 1.57$. Ask it to explain why adding demand-group indicators changes the comparison. A numerical match without that explanation is an incomplete interpretation.

For a larger dataset, specify the formula and desired covariance estimator before requesting code. Ask the assistant to list dropped rows and explain how categorical variables were encoded. Have it calculate a predicted contrast at two values of a moderator and compare the result with the interaction formula. Treat diagnostic recommendations as hypotheses to inspect, rather than automatic permission to remove observations or alter the model.

For practice, consider an advertising coefficient measured per $1,000 of spending. Re-express the predictor in dollars and explain how the numerical coefficient changes while fitted sales remain the same. Next explain why a robust standard error cannot eliminate the bias in the market-demand example. Finally, distinguish a confidence interval for the conditional mean from a prediction interval for a new store's realized sales: the latter also reflects individual outcome variation.

3.9 Readings

An Introduction to Statistical Learning provides a foundation for regression, model assessment, and nonlinear extensions. The statsmodels regression documentation supports implementation.

Chapter 4

Product Preferences, Customer Segments, and Lifetime Value

A subscription retailer must choose a product bundle, decide whom to serve, and determine how much to spend acquiring or retaining customers. These decisions are connected. A feature that attracts one segment may be costly to deliver, and a segment with high initial spending may have low retention. Conjoint analysis estimates tradeoffs among product attributes, clustering summarizes patterns of similarity, and customer lifetime value translates future behavior into expected contribution.

4.1 Product Design Requires Tradeoffs

Asking consumers whether they like fast delivery often produces a favorable response, but it does not reveal how much they will pay for it or which other feature they will give up. Conjoint tasks present product profiles whose attributes vary. Respondents may rate profiles, rank them, or choose among alternatives. Choice-based tasks resemble an assortment decision, while ratings can provide more observations per respondent under a different response process.

Attributes should represent decisions the firm can implement and descriptions respondents can understand. Levels should span a plausible range without creating impossible combinations. If every premium-quality product also has the highest price, the design provides little independent information about the quality-price tradeoff. Randomization and a suitable experimental design help separate attribute effects, but realistic constraints may limit which combinations can be shown.

Include the decision's relevant alternatives. A forced choice between two products estimates preference conditional on choosing one of them. An outside option such as "buy neither" helps represent category participation. A simulation that omits the outside option can overstate category demand because every respondent must buy something. Survey design, task burden, and hypothetical behavior also affect how well stated choices transfer to market behavior.

4.2 Utility, Choice Probabilities, and Willingness to Pay

A simple model assigns respondent i a systematic utility for alternative j:

$$V_{ij} = \beta_q q_j + \beta_f f_j + \beta_p p_j,$$

where q_j indicates premium quality, f_j indicates a delivery feature, and p_j is price in dollars. Utility has an arbitrary scale. Coefficients describe tradeoffs on that scale, and price is expected to enter negatively in a conventional demand interpretation. Alternative-specific constants can capture average preference for named options relative to a reference alternative.

Under a multinomial-logit random-utility specification, the probability of choosing j from the available set $\mathcal{J}_i$ is

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_{k \in \mathcal{J}_i} \exp(V_{ik})}.$$

Multinomial logit imposes proportional substitution across alternatives. Whether this pattern fits the product category requires empirical assessment. Repeated choices by one respondent are dependent; estimation and validation should respect that respondent structure. Heterogeneous preferences may be represented with interactions, latent classes, or random coefficients when the data support the additional complexity.

For a binary feature whose coefficient is β_f, a local willingness-to-pay tradeoff under linear price utility is $-\beta_f/\beta_p$. If $\beta_f = 1.2$ and $\beta_p = -0.1$ per dollar, the model equates the feature with \$12 of price. This ratio is conditional on the specification and utility scale cancellation. A price coefficient close to zero makes the ratio unstable, and preferences outside the experimental price range are extrapolations.

Attribute importance is often calculated from the range of an attribute's estimated part-worths relative to the sum of ranges. The result depends on the levels included in the study. Expanding the price range can increase measured price importance even when underlying preferences are unchanged. Report the design range when discussing importance.

4.3 Case Study: A New Bundle Can Cannibalize Existing Sales

Consider two subscription bundles and an outside option. Bundle A has premium quality and costs \$10; bundle B has standard quality and costs \$6. Let $\beta_q = 1.2$, $\beta_p = -0.1$, and normalize outside-option utility to zero. The systematic utilities are 0.2 for A and -0.6 for B. The logit probabilities are approximately 44.1% for A, 19.8% for B, and 36.1% for the outside option.

Now introduce bundle C with the same modeled attributes as A. The logit model assigns A and C approximately 30.6% each, B 13.7%, and the outside option 25.1%. C's share is 30.6 percentage points, but the total purchasing share rises only from 63.9% to 74.9%, an increase of about 11.0 points. The remaining share comes from the existing products. Forecasting a new product's sales without accounting for substitution would overstate incremental demand.

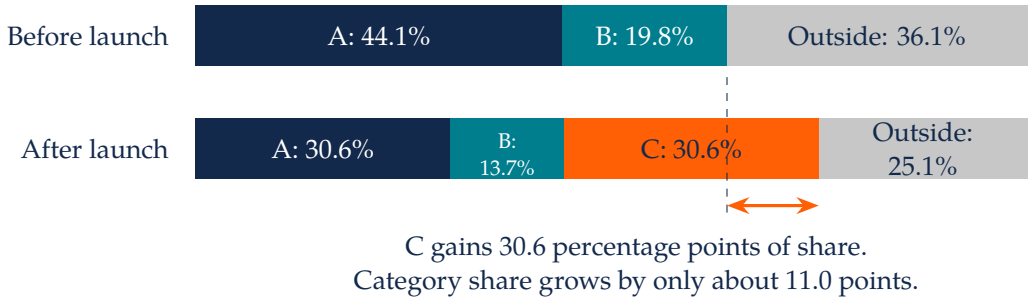


Figure 4.1. A new product's sales include substitution from existing products.

With utility coefficients and the potential-customer population fixed, adding C changes only the available choice set. Because C has the same modeled attributes as A, it is a close substitute. The logit model's independence-of-irrelevant-alternatives property makes C draw share proportionally from all existing alternatives. Whether that pattern is plausible is an empirical question. Richer substitution models may be needed when close substitutes compete especially strongly with one another.

A profit calculation must attach contribution margins and fixed costs to the simulated quantities. If A and C each contribute $4 per purchase and B contributes $3, expected contribution per potential customer rises from about $2.36 to $2.86 before product-development and operating costs. The contribution comparison is conditional on the specified utilities, margins, and fixed customer population. A pilot or holdout choice tasks can help evaluate the proposed product and model before a larger decision.

4.4 Estimate and Validate Preferences

A conjoint workflow begins with the managerial choice and feasible attributes, then constructs the task design and recruits a relevant respondent sample. Inspect completion time, straight-line responses, inconsistent attention, and whether participants understood the attributes. Exclusion rules should be set and reported transparently. Removing respondents because their preferences are inconvenient changes the target population.

Estimate a transparent baseline before adopting a more elaborate model. Check coefficient coding, reference levels, price units, and whether probabilities sum to one for every choice set. Evaluate prediction on held-out tasks or respondents according to the intended use. A model that fits familiar respondents' repeated choices may generalize differently to a new customer population.

A useful simulation changes one decision at a time and records what is fixed: attributes, price, competitor options, market size, and margins. Share of preference from a survey is not automatically market share. Distribution, awareness, availability, purchase frequency, and sampling can create differences. State the link between the conjoint output and the demand quantity used in the business calculation.

4.5 Segmentation Begins With a Definition of Similarity

A retailer may want segments for differentiated service, product positioning, or campaign design. Those objectives should guide the variables used to define similarity. Grouping customers by recent spending answers a different question from grouping them by category preferences or service needs. A segment should be interpretable, sufficiently stable, and connected to an action the firm can take.

Recency, frequency, and monetary value provide a useful descriptive starting point. Define recency as time since the last eligible purchase at a common observation date, frequency as the number of eligible purchases during a fixed window, and monetary value as sales or contribution during that window. Customers with different observation histories require adjustment or a deliberate eligibility rule. A new customer observed for one month should not be compared mechanically with one observed for a year.

K-means selects K centers to minimize within-cluster squared Euclidean distance:

$$\sum_{i=1}^{n} \left\| z_i - \mu_{c(i)} \right\|^2 .$$

Here z_i is the vector of transformed customer features, $c(i)$ is customer i's assigned cluster, and $\mu_{c(i)}$ is its cluster center. The feature transformations determine the distance. Standardization often helps prevent a large numerical scale from dominating, but it does not establish that all features should have equal managerial importance.

For three customers with spending-frequency pairs A=(100,1), B=(110,9), and C=(200,1), raw Euclidean distance makes B much closer to A than C is. If spending is expressed in hundreds of dollars while frequency remains in purchases, C becomes closer to A. The customer records have not changed; the implicit weights have. This is why units and scaling belong in the substantive definition of a segment.

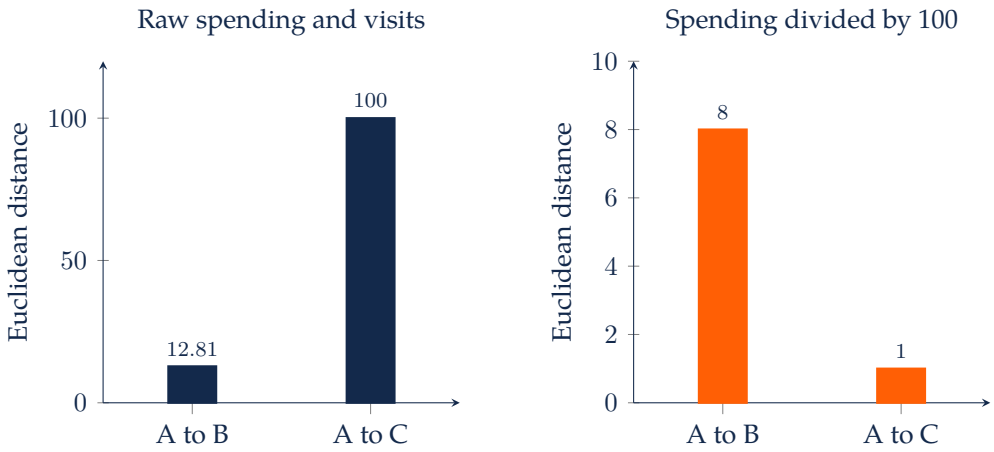


Figure 4.2. Changing feature scales changes which customers appear similar.

Changing spending units reverses the nearest neighbor because it changes the relative weight placed on spending and visits. A segmentation report should explain that metric and examine whether the proposed actions remain sensible under reasonable transformations.

4.6 Choosing and Evaluating Segments

Use more than one criterion to choose K. Within-cluster dispersion falls mechanically as more clusters are added. A silhouette measure compares cohesion with separation under the chosen distance, but a high silhouette does not establish business value. Examine segment sizes, profiles, stability across resamples or time, and whether the segments imply different feasible actions.

Hierarchical clustering provides a nested set of groupings whose interpretation depends on the linkage and distance. Density-based methods can identify irregular groups and mark some observations as noise, with sensitivity to neighborhood parameters. Mixture models provide probabilistic membership under distributional assumptions. These alternatives address different patterns;

no algorithm discovers a uniquely correct segmentation independent of representation and purpose.

For customer activation, avoid using future purchase outcomes to define a segment that supposedly exists before the campaign. For evaluating segmentation, examine behavior not used mechanically to construct it. A high-spending segment will have high spending by design if spending is a clustering input. A more useful question is whether the grouping helps predict future needs or improves the result of a tested service policy.

Clustering shopping trips on basket size, category composition, and spending can help separate trip types before comparing channels. Fit the representation without allowing the desired channel comparison to determine the grouping. Examine profiles and sensitivity, because a trip-classification algorithm can also create misleading comparisons if its features or thresholds differ across channels.

4.7 Customer Lifetime Value Depends on Timing and Margin

Customer lifetime value is the discounted expected contribution from a relationship, under a stated time horizon and behavioral model. For a newly acquired customer, a simple finite-horizon expression is

$$\text{CLV} = \sum_{t=1}^{T} \frac{\Pr(\text{active at } t)\, m_t}{(1+d)^t} - a.$$

Here t indexes periods after acquisition, m_t is expected contribution conditional on being active in period t, d is the discount rate per period, and a is acquisition cost paid at time zero. Contribution should deduct relevant variable service and fulfillment costs. The expression assumes the conditional margin and activity probability have been defined consistently; more general models can use the expected product of activity and customer-specific margin.

Assume a stationary subscription process: a currently active customer survives each period with probability r, earns a constant contribution m at the end of a period only if still active, and has an infinite horizon. With $r/(1+d) < 1$, the geometric series gives

$$\text{CLV} = \frac{mr}{1+d-r} - a.$$

This convention excludes an immediate current-period contribution. Including one would add that contribution and change the formula. Monthly retention requires monthly margins and a monthly discount rate. Combining monthly retention with an annual discount rate produces an inconsistent calculation.

With monthly contribution $m = 60$, retention $r = 0.80$, monthly discount rate $d = 0.01$, and acquisition cost $a = 50$, CLV is approximately \$178.57. Increasing retention to 0.90 raises it to approximately \$440.91. The gain is nonlinear because retention affects many future payments. These values are conditional on constant retention and margin; changes in retention or margin over time would change the valuation.

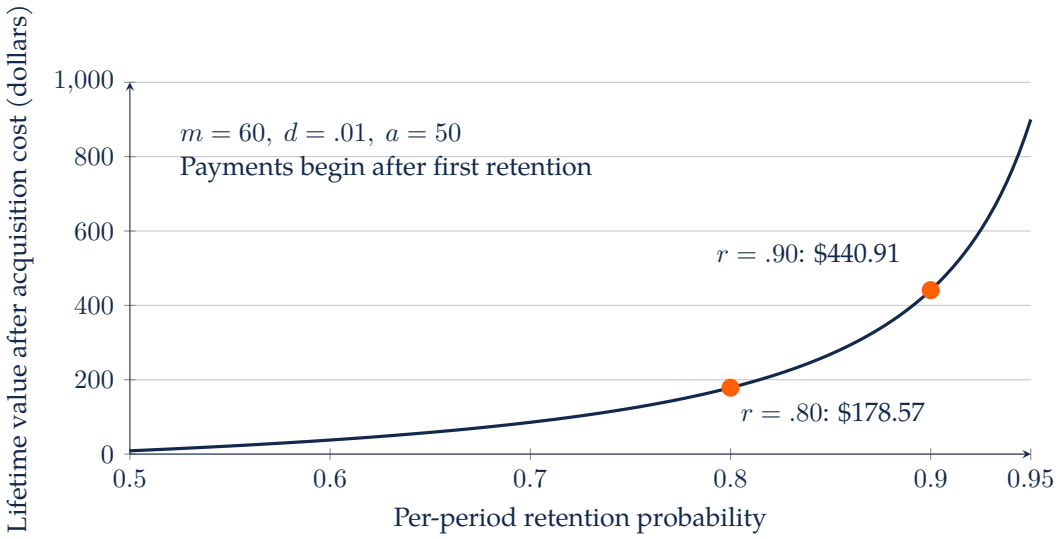


Figure 4.3. The value of retention is nonlinear under a stationary subscription model.

With monthly contribution, discounting, and acquisition cost fixed, a one-point retention increase has greater value at a higher baseline retention rate. Uncertainty about long-run retention can therefore dominate a CLV forecast.

4.8 Subscription and Noncontractual Relationships

In a subscription setting, cancellation is often observed. Survival analysis can model time to cancellation and account for right censoring when observation ends before cancellation. A hazard describes the probability of exit during a period conditional on survival to that period. Cohort effects, tenure dependence, and changes in service quality can make constant retention implausible.

In a noncontractual retail setting, the last purchase does not reveal whether a customer has permanently left. Models such as BG/NBD represent purchase frequency and latent dropout under specific stochastic assumptions. A separate spending model may estimate value per transaction. Those assumptions require checking, including the treatment of returns and any assumed relationship between purchase frequency and monetary value. A customer-level count alone does not identify permanent churn without a model.

Evaluate value forecasts on a holdout period with the same horizon used in the decision. Compare aggregate and customer-level calibration, rank ordering, and errors for new or infrequent customers. A model can rank customers well while overpredicting total value. For an acquisition budget, the aggregate monetary error may be more consequential than a small improvement in ranking.

4.9 Integrated Case: Which Segment Should Receive Retention Support?

Suppose segment H has high predicted CLV but little response to service outreach, while segment M has moderate predicted CLV and a larger response. Targeting only H can be inferior even if its value forecast is accurate. The relevant quantity is the change in expected future contribution caused by outreach, net of the outreach cost. CLV describes a baseline relationship; a retention intervention changes that relationship under a separate causal model.

A useful analysis would estimate or specify segment-specific preferences, margins, and retention, then test the proposed offer. Product design might attract a segment with expensive service needs; a seemingly attractive acquisition price might therefore have low contribution. Segment labels should help organize the decision without replacing these calculations. A final recommendation should show which assumption changes the preferred segment or product.

For practice, recompute the subscription value with $r = 0.85$ and verify the result before acquisition cost is subtracted. Add an immediate \$60 contribution and explain the change in timing. In the conjoint example, remove the outside option and observe that simulated purchase incidence becomes one by construction. In the clustering example, compare similarity using category shares instead of spending levels and explain which managerial question the new representation serves.

4.10 AI-Assisted Implementation and Readings

Ask a coding assistant to calculate choice probabilities from a supplied profile table, check that they sum to one, and separate a new product's share from incremental category demand. For clustering, require documented transformations, segment profiles, and a sensitivity comparison. For CLV, request an explicit payment timeline and verify the closed-form calculation against a long finite sum. These checks catch common mistakes that can survive apparently reasonable charts.

Sawtooth's academic materials provide conjoint examples. Dolnicar, Grün, and Leisch's segmentation book develops the connection between segmentation design and use. PyMC-Marketing's CLV documentation offers implementation examples whose model assumptions and software versions should be checked before adaptation.

Chapter 5

Text, Social Media, and Customer Networks

A retailer receives thousands of reviews and support messages each week. The service manager needs to distinguish delivery delays, product defects, and billing problems, then decide which issues deserve attention. A count of negative words is an incomplete representation: "the refund was fast, but the replacement failed" contains different evaluations of two aspects of the experience. The analytical task is to turn language into a defensible measurement of a concept relevant to action.

Text analysis also raises a sampling problem. Customers who post reviews can differ from those who remain silent, and the platform can affect what gets posted or displayed. Social-network analysis adds another layer: observed connections and similar behavior can reflect influence, common circumstances, or selection into relationships.

5.1 Define the Construct and the Document

A construct is the concept the analysis intends to measure, such as delivery dissatisfaction, perceived authenticity, or purchase intent. A label is the operational category assigned to an observation. A document might be a complete review, one sentence, a support ticket, or a conversation. Those units should follow the construct. A review-level sentiment label can obscure a positive service experience and a negative product experience in the same review.

Write an annotation codebook before producing labels at scale. It should define each category, inclusion and exclusion rules, treatment of ambiguity, and examples near category boundaries. If a message can contain several problems, a multi-label task may be appropriate. If the decision requires routing each ticket to one team, define a primary issue or a routing priority separately from the presence of every issue.

Metadata can be informative and potentially misleading. Ratings may help validate a sentiment measure but can also leak the intended label into a prediction exercise. Product category, language, channel, and date can reveal domain differences. Remove or protect personal information before sending text to an external service, and use approved data access. A scraped review is not automatically free of personal or contractual restrictions.

5.2 Representing Language With Counts and Weights

A bag-of-words representation counts tokens while largely ignoring their order. An n-gram representation retains local sequences, so "not good" can differ from "good." Tokenization, case normalization, punctuation, and stop-word rules change the resulting features. Removing negation words can destroy information needed for sentiment. Removing product names may help focus on attributes or may erase the very object whose complaints the manager needs to identify.

Term frequency-inverse document frequency gives greater weight to terms that are frequent in a document but less common across the corpus. One simple convention is

$$\text{tfidf}_{wd} = \text{tf}_{wd} \log\left(\frac{N}{\text{df}_w}\right),$$

where tf_{wd} is the frequency of word w in document d, N is the number of documents, and df_w is the number of documents containing the word. Implementations vary in smoothing, normalization, and term-frequency definitions. Record the convention rather than assuming a library's output matches this exact expression.

If "delivery" appears in two of three documents, its inverse-document-frequency term under this convention is $\log(3/2) \approx 0.405$. A word appearing in every document receives zero weight. That weighting can help discriminate documents, but a common term may still be substantively important. Statistical distinctiveness and managerial importance are different criteria.

Fit a learned vocabulary and IDF weights on training text when evaluating prediction. Learning them from the test corpus exposes information about the evaluation distribution. The leakage may appear minor compared with using labels, yet it still changes the task. A pipeline makes the training-only transformation explicit and reproducible.

5.3 Embeddings and Semantic Similarity

An embedding maps a document or token into a vector learned from language data. Similar vectors can capture related meanings even when documents use different words. Cosine similarity compares the angle between vectors:

$$\cos(u, v) = \frac{u^\top v}{\|u\| \, \|v\|}.$$

The normalization removes vector length from the comparison. This is useful when direction represents meaning in the selected model, but it does not guarantee that the similarity corresponds to the marketing construct. Two messages about refunds can be semantically close while expressing opposite evaluations. Retrieval quality should be checked with examples that matter for the downstream task.

A simple TF-IDF classifier can be a strong baseline for a well-defined domain. Embeddings may help with paraphrases, sparse labels, and semantic retrieval, but they introduce dependence on a pretrained model and its training domain. Compare alternatives on the same held-out data and operational criteria. Record model versions because an updated representation can change nearest neighbors and labels even when the input text is unchanged.

For interpretation, inspect representative successes and failures. A model may associate a brand name with complaints because one product had a recall during training. It can then misclassify neutral mentions after the issue is resolved. Temporal validation and product-level holdouts help reveal whether performance depends on such shortcuts.

5.4 Sentiment, Aspect Labels, and Topic Models

A sentiment lexicon assigns scores to words and combines them using rules. It offers a transparent baseline, but sarcasm, domain-specific meanings, and context can cause errors. A supervised classifier learns a mapping from labeled examples to outcomes. Its labels and training sample define what it learns; a generic "positive/negative" dataset may not represent customer-service severity.

Aspect-based analysis separates the object being evaluated from the evaluation itself. For the sentence "delivery was late, but the support agent solved the problem," a useful record might contain delivery-negative and service-positive labels. A single overall score discards that distinction. If the action is a logistics intervention, the aspect-specific measure is closer to the decision.

Topic models describe recurring patterns of word co-occurrence or document similarity. A topic is an estimated pattern, not automatically a stable consumer need. In probabilistic models, documents can combine topics; in clustering approaches, assignments may be discrete or probabilistic. Analysts supply topic names after inspecting high-weight words and representative documents. Validate those names with actual text and examine stability across preprocessing and model choices.

A manager should avoid interpreting an increase in a topic's share without checking its denominator. More delivery complaints can mean more delivery problems, more deliveries overall, a changed review prompt, or fewer reviews about other topics. Combining text measures with transaction or service-volume data helps distinguish these explanations.

5.5 Case Study: A Complaint Detector Under Changing Prevalence

Suppose a classifier identifies a rare but costly product defect. In a synthetic evaluation population of 1,000 tickets, 200 truly mention the defect. The classifier detects 160 of them and incorrectly flags 80 of the 800 other tickets. Sensitivity, or recall, is 80%; specificity is 90%; precision is 160/240, or 66.7%.

Population	True Defect Tickets	True Positives	False Positives	Precision
Evaluation sample	200	160	80	66.7%
Deployment sample	10	8	99	7.5%

The second row holds sensitivity at 80% and specificity at 90%, but reduces true prevalence to 1%. Among 990 nondefect tickets, 99 are flagged. Only eight of the 107 flagged tickets are true positives. A manager using the original precision to plan review staffing would be badly misled.

Bayes' rule makes the dependence explicit. If prevalence is ρ, sensitivity is s, and specificity is q, then

$$\Pr(\text{defect} \mid \text{flag}) = \frac{s\rho}{s\rho + (1-q)(1-\rho)}.$$

The denominator counts both correct and incorrect flags. For a rare issue, even a modest false-positive rate can dominate that total. This is why validation should represent the deployment population or explicitly adjust for a changed sampling design.

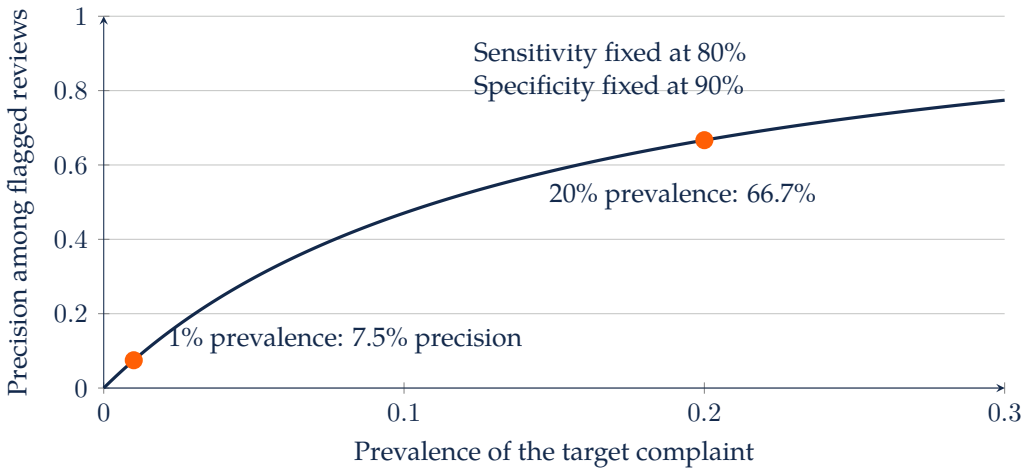


Figure 5.1. Precision falls as the target issue becomes rarer, even with unchanged sensitivity and specificity.

Sensitivity and specificity remain at 0.80 and 0.90 in both populations. The decline in precision comes entirely from the lower prevalence. Operational evaluation should also consider the cost of missed defects, the burden of false alarms, and whether uncertain cases can be routed to human review.

5.6 Human and LLM Annotation

LLMs can help draft a codebook, suggest candidate labels, or classify text using explicit instructions. For a defensible measurement exercise, preserve the original text, the codebook version, model identifier, prompt, output schema, and date. Require a bounded output format with an allowed "uncertain" category where appropriate. Validate the output against the schema and inspect cases that violate the instructions.

A prompt should identify the construct, unit, categories, boundary rules, and examples. For delivery dissatisfaction, it should distinguish a complaint about an actual delay from a question about the promised delivery date. Ask for a short supporting span from the supplied text when that improves auditability. A fluent explanation is not evidence that the label is correct; compare it with independent human judgments on a relevant sample.

Human agreement is informative because the construct may itself be ambiguous. Double-code a sample, examine disagreements, and refine definitions before scaling. Report category-specific errors, prevalence, and performance across languages or product groups. An average accuracy can conceal systematic failure on the minority category that motivates the analysis.

Treat text being classified as data. A review may contain language resembling an instruction to the model. It should not change the annotation task, request external actions, or supply new system rules. Separate the fixed task instructions from the document content, constrain outputs, and inspect unexpected behavior. Use synthetic or approved text for classroom exercises involving external services.

5.7 Networks: Connections, Reach, and Brokerage

A network represents entities as nodes and relationships as edges. Nodes might be customers, creators, brands, or products. An edge might indicate following, communication, co-purchase, or a similarity criterion. Directed and weighted networks preserve different information from undirected binary networks. The edge definition should follow the substantive relationship the analyst intends to study.

Degree counts a node's direct connections. Betweenness summarizes how often a node lies on shortest paths between other nodes. A person with few direct contacts can have high betweenness if those contacts connect otherwise separate groups. Eigenvector-based measures give more weight to connections with well-connected nodes. These metrics describe network structure under their definitions; they do not by themselves establish persuasion, credibility, or the effect of paying a creator.

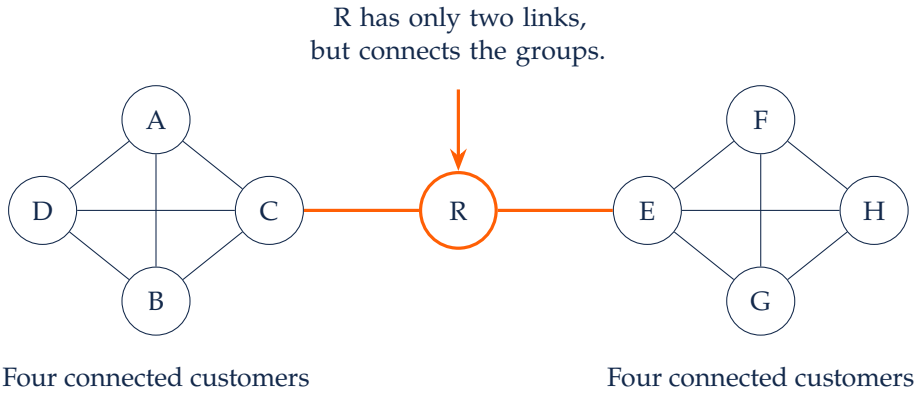


Figure 5.2. A broker with few direct links can connect two otherwise separate customer groups.

Broker R has two direct connections, while each within-group connector has four. Removing R disconnects the two groups. A customer with fewer direct contacts can therefore be essential for reaching a different audience. A marketing campaign may value either role, depending on message transmission, audience overlap, and the intended outcome.

Observed similarity among connected customers can arise from influence, homophily, or common exposure. Friends may adopt a product because they affect one another, because they already share tastes, or because they see the same promotion. Regressing adoption on friends' adoption does not distinguish those mechanisms without additional design. Randomized encouragement, timing restrictions, or credible natural variation can help, with attention to spillovers and the unit of assignment.

5.8 Social-Media Case: Engagement and Commercial Value

A brand compares two creators. One receives many comments, but those comments largely come from an audience that already follows the brand. The other reaches a smaller group with less overlap. A high engagement count does not establish more incremental sales. The comparison

requires a target population, exposure definition, outcome horizon, and a way to measure what would have happened without the collaboration.

Text can help describe comment content, and network overlap can describe audience duplication. Those measurements support campaign design, but they should be combined with an evaluation strategy. A test might randomize eligible regions or audience groups when feasible, or compare outcomes using a credible observational design. If communication crosses groups, spillovers can complicate the interpretation of an untreated group.

For classroom practice, construct two small audience sets and calculate their overlap. Then compare comments per exposed user with total comments. Explain why neither is a direct estimate of incremental profit.

5.9 AI-Assisted Workflow, Practice, and Readings

Start a text exercise with a small manually inspected sample and a codebook. Build a transparent baseline, reserve an evaluation sample, and compare a more complex method on the same outcomes. Ask an assistant to produce a confusion matrix and representative errors, then independently recompute precision and recall from the counts. Keep model selection and prompt refinement away from the final test sample.

For practice, calculate accuracy in the first complaint-detector sample: 88%. Explain why a classifier that always predicts "no defect" would achieve 99% accuracy in the rare-prevalence deployment sample while detecting no defects. Next identify how removing "not" changes the meaning of "not reliable." Finally, explain why two highly connected adopters may share preferences even if neither caused the other's adoption.

The scikit-learn text-feature documentation supports count and TF-IDF representations. VADER's author repository provides a rule-based sentiment baseline. Hagen and colleagues (2020) discuss ways machine learning can support behavioral marketing research.

Chapter 6

Causal Effects and Marketing Experiments

A retailer wants to know whether an email campaign increases contribution. Customers who open marketing emails often purchase more, but they may already be more engaged. Comparing openers with nonopeners can therefore reward a campaign for behavior it did not cause. The decision requires a comparison between outcomes under alternative actions for the same target population.

6.1 Define the Treatment, Outcome, and Estimand

Let $Y_i(1)$ be customer i's 30-day contribution if assigned an email and $Y_i(0)$ the contribution if not assigned. The individual effect is $Y_i(1) - Y_i(0)$. Only one of these outcomes is observed for a given customer under a single assignment. A causal design supplies a defensible way to learn an average comparison despite that missing counterfactual.

For a target population, the average treatment effect is

$$\text{ATE} = E[Y_i(1) - Y_i(0)].$$

The expectation refers to the stated population and outcome window. An average treatment effect on the treated uses the population that actually received treatment. An intention-to-treat effect uses assignment as the treatment, even if some assigned customers never open the email. These are different estimands. The managerial decision to send an email often corresponds closely to the assignment effect, because opening is outside the firm's direct control.

A well-defined treatment includes the offer, timing, channel, and implementation. "Digital marketing" is too broad to identify a single comparison. The outcome should include economically relevant costs and have enough follow-up to capture returns or delayed purchases. If customers can affect one another, an individual's outcome may depend on other customers' assignments. The unit and design then need to account for interference rather than silently assuming it away.

6.2 Why the Treated-Untreated Difference Can Be Biased

The observed difference between treated and untreated customers can be decomposed as

$$E[Y \mid D = 1] - E[Y \mid D = 0] = E[Y(1) - Y(0) \mid D = 1] + \{E[Y(0) \mid D = 1] - E[Y(0) \mid D = 0]\}.$$

Here D indicates treatment. The first term is the average treatment effect among treated customers. The bracketed term compares their untreated outcomes with untreated customers' outcomes. If the firm targets people who would purchase anyway, that selection term can be positive. If it targets customers in serious decline, the term can be negative. The direction follows the assignment process, not a universal rule about observational data.

An unobserved engagement variable may affect both email opening and purchasing. Adjusting for pre-treatment engagement measures can be useful if they capture the relevant confounding, but controlling for a variable does not guarantee this condition. A post-treatment variable such as clicking the email can mediate part of its effect. Conditioning on customers who click changes the population and can introduce selection.

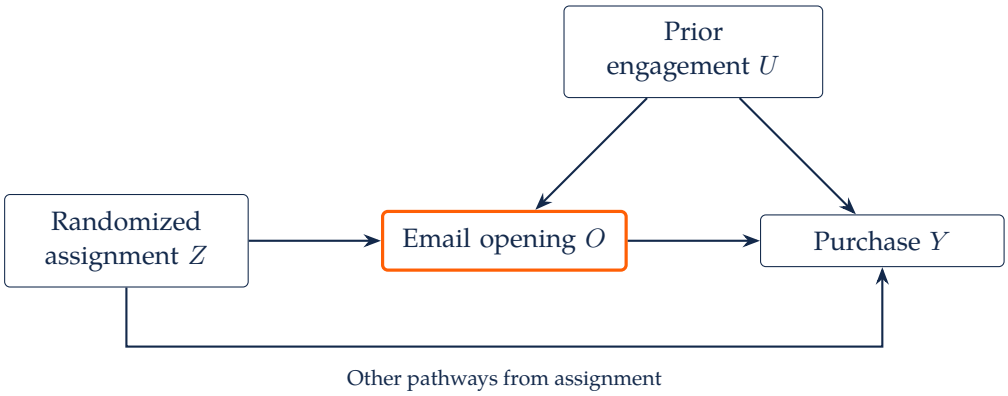


Figure 6.1. Assignment, engagement, and opening define different causal comparisons.

The firm randomizes email assignment, while customers choose whether to open. Assignment can affect opening, and engagement can affect both opening and purchases. Restricting attention to openers creates a selected group. An assignment-based analysis preserves the randomized comparison when its implementation and outcome measurement are valid.

6.3 Designing an Experiment Around the Decision

An experiment needs an eligible population, assignment unit, treatment implementation, primary outcome, horizon, and analysis plan. Randomize at a unit that corresponds to exposure and likely spillovers. A customer-level email experiment may be appropriate when accounts act independently. A store-level display intervention requires store assignment or a design that accounts for shared exposure. Household, geographic, or network spillovers can require broader clusters.

Balance treatment groups through randomization, with blocking or stratification when important pre-treatment characteristics are known. For example, randomize within customer-tenure groups to improve balance and precision. Record assignment before contact and retain every assigned unit in the primary analysis when estimating the intention-to-treat effect. Excluding nonopeners, customers who later unsubscribe, or stores with disappointing execution can undermine the randomized comparison.

Specify a primary outcome and guardrails before viewing results. Incremental contribution might be primary, with unsubscribe rates, complaints, and returns as guardrails. Multiple outcomes can be informative, but selecting the most favorable result after examining many tests exaggerates evidence. Decide how multiple testing, sequential monitoring, and stopping will be

handled. Ordinary fixed-horizon p-values do not generally retain their advertised interpretation under unrestricted repeated peeking and outcome selection.

6.4 Case Study: Estimate an Email Assignment Effect

In a synthetic experiment, 1,000 customers are assigned an email and 1,000 are assigned no email. Within 30 days, 130 assigned customers and 100 controls make at least one eligible purchase. The estimated assignment effect on purchase incidence is

$$\hat{\tau} = \frac{130}{1000} - \frac{100}{1000} = 0.03,$$

or three percentage points. Relative to the control rate of 10%, this is a 30% relative increase. The two descriptions use different denominators; reporting only "30% lift" can obscure the absolute change.

Under independent customer assignments and the usual large-sample approximation, an unpooled standard error is

$$\widehat{\mathrm{SE}}(\hat{\tau}) = \sqrt{\frac{0.13(0.87)}{1000} + \frac{0.10(0.90)}{1000}} \approx 0.0143.$$

An approximate 95% interval is $[0.0021, 0.0579]$, or about 0.21 to 5.79 percentage points. This interval describes sampling uncertainty under the stated design and approximation. It does not account for an unmeasured outcome-definition error or interference between assigned customers.

Suppose an additional purchase contributes \$40 and email delivery costs \$0.50 per assigned customer. If the offer has no additional discount cost and purchase incidence is the relevant incremental quantity, estimated incremental contribution is $\$40(0.03)$-\$0.50=\$0.70 per assigned customer. Applying the same transformation to the effect interval gives approximately \$-0.42 to \$1.82. A positive point estimate therefore coexists with economically consequential uncertainty. A decision can combine that uncertainty with rollout costs, the value of another test, and guardrail outcomes.

The margin assumption matters. If treatment also changes basket size, returns, or discount expense, purchase incidence alone is insufficient. An experiment measuring contribution directly may better match the objective.

6.5 Power, Precision, and the Minimum Useful Effect

Power planning begins with the smallest effect that could change the decision, not with a convenient number of rows. For two equally sized independent groups and a binary outcome with baseline rate near p, a rough per-group sample-size approximation is

$$n \approx \frac{2p(1-p)(z_{1-\alpha/2} + z_{1-\beta})^2}{\delta^2},$$

where δ is the absolute effect in probability units, α is the two-sided significance level, and $1-\beta$ is target power. This approximation uses a common baseline variance and normal critical values.

More precise planning can use the expected rates under the alternative and the actual assignment design.

With $p = 0.10$, $\delta = 0.02$, $\alpha = 0.05$, and 80% power, using 1.96 and 0.84 gives about 3,528 customers per group. Halving the target effect approximately quadruples the required sample under this formula. A two-percentage-point effect is 0.02, not 2; a unit error can produce a meaningless sample-size recommendation.

Clustering reduces the amount of independent information. A common approximation with equal cluster size m and intracluster correlation ρ multiplies individual-level variance by $1 + (m - 1)\rho$. For $m = 20$ and $\rho = 0.05$, that factor is 1.95. The number of clusters, unequal cluster sizes, assignment design, and small-sample inference still require explicit consideration. Thousands of customers across four stores do not provide thousands of independent store-level treatment assignments.

6.6 Implementation Checks That Protect the Comparison

Verify that the assignment counts match the intended allocation and investigate material sample-ratio discrepancies. Check whether tracking differs between treatment and control, whether treatment changes the probability an outcome is observed, and whether exclusions were applied consistently. Pre-treatment balance is a diagnostic, not proof that every causal assumption holds. In a correctly randomized experiment, chance imbalances can occur.

Noncompliance changes the estimand. If only 40% of assigned customers open the email, the intention-to-treat effect remains the effect of assignment under the observed implementation. Dividing it by an opening-rate difference identifies a complier effect only under additional instrumental-variable conditions, including an appropriate exclusion restriction and monotonicity. The firm may care about the assignment effect even when a different scientific question concerns actual exposure.

Use pre-treatment outcomes to improve precision when justified. A regression adjustment can compare outcomes while accounting for baseline spending or stratification variables. Avoid selecting an adjustment because it makes the treatment coefficient significant. Report the planned analysis and explain any departures. Analyze heterogeneity cautiously, especially when many subgroups are examined after seeing the main result.

6.7 Observational Adjustment and Its Limits

When randomization is unavailable, a common approach assumes that treatment is conditionally independent of potential outcomes given observed pre-treatment covariates, with overlap in treatment probabilities. Regression adjustment, matching, and inverse-probability weighting implement different ways to construct a comparison under that condition. The central question is whether the observed covariates make the treated and untreated outcomes comparable.

A propensity score is the treatment probability given those covariates. Extreme estimated scores indicate limited overlap and can create unstable weights. Trimming or restricting the sample changes the population for which the effect is estimated. Report that population explicitly. A flex-

ible prediction algorithm for treatment assignment does not establish that all confounders were measured.

A doubly robust estimator combines an outcome model and a treatment model and can be consistent when one nuisance model is correctly specified, under the underlying identification conditions and regularity assumptions. This robustness does not cover unmeasured confounding. For advanced implementations, sample splitting and cross-fitting help control overfitting of nuisance functions; the design remains the foundation of the causal interpretation.

6.8 Difference-in-Differences and a Counterfactual Trend

A retailer introduces an app in treated markets while comparable markets remain untreated. A before-after difference in treated markets mixes the app's effect with common demand changes. Difference-in-differences subtracts the comparison group's change:

$$\hat{\tau}_{\text{DiD}} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre}).$$

In a synthetic example, treated-market sales rise from 120 to 150 and comparison-market sales rise from 100 to 110. The DiD estimate is 20 sales units. Under the identifying parallel-trends condition, the treated markets' untreated counterfactual would have risen to 130. Different starting levels are compatible with this design; the condition concerns untreated changes.

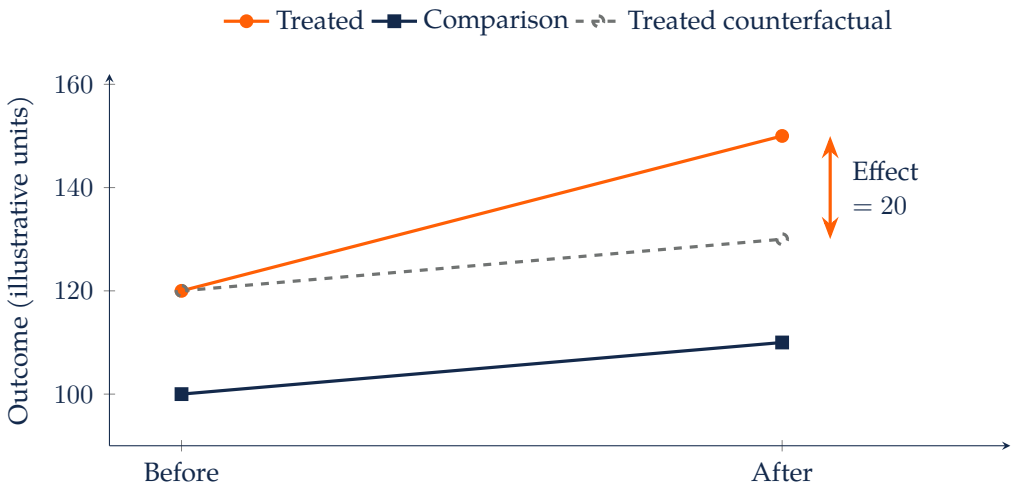


Figure 6.2. Difference-in-differences subtracts the comparison group's change to construct the counterfactual.

The dashed line is a counterfactual implied by parallel untreated trends, not an observed sales series. The vertical gap between observed treated sales and that line is the estimated effect. A treatment-specific local shock at the same time would challenge the interpretation even if the pre-treatment levels looked similar.

Inspect multiple pre-treatment periods, institutional timing, possible anticipation, spillovers, and changes in sample composition. Similar pre-trends can support the design's plausibility, but a nonsignificant pre-trend test does not establish parallel counterfactual trends. With staggered

adoption and heterogeneous effects, conventional two-way fixed-effects coefficients can combine comparisons in problematic ways. Use estimators aligned with cohort and time-specific effects, with a clear comparison group, when extending beyond the simple two-period design.

6.9 Other Designs: Discontinuities and Instruments

A regression discontinuity design can use a treatment rule at a threshold, such as eligibility based on a pre-existing score. Customers just above and below the cutoff may be comparable if other determinants of the outcome vary smoothly there and the score is not precisely manipulated. The resulting effect is local to the cutoff and the relevant compliance behavior. Bandwidth, functional form, density patterns, and discontinuities in other predetermined variables deserve inspection.

An instrumental variable changes treatment while affecting the outcome only through the treatment channel, under the design's conditions. Relevance is an empirical first-stage requirement; exclusion is a substantive condition. A strong first stage cannot establish exclusion. In a binary-treatment setting with independence, exclusion, and monotonicity, the familiar local average treatment effect concerns compliers whose treatment changes with the instrument. It need not equal the population ATE.

Marketing instruments require particularly careful reasoning because pricing, distribution, and exposure decisions often respond to demand. A variable correlated with treatment is not sufficient. Explain who makes the relevant decision, when the instrument is determined, and which other channels could connect it to outcomes. If those channels remain credible, the causal interpretation needs revision.

6.10 Spillovers and Experiment Units

An advertising campaign can shift purchases between stores or products rather than increase total demand. A customer-level outcome may show a gain while chain-wide contribution changes little. Define the outcome at the level relevant to the managerial objective and examine displacement where it is plausible. The same issue arises when customers share offers or when a platform changes recommendations for multiple users.

An assigned store can gain sales while diverting purchases from another store. If the comparison store loses customers, subtracting its decline can overstate the increase in total chain sales. Cluster randomization, geographic separation, or designs that vary treatment saturation may help identify the relevant effects, subject to their own practical constraints.

6.11 AI-Assisted Workflow and Practice

Ask an assistant to state the estimand, assignment unit, outcome horizon, and comparison group before writing estimation code. For the email example, require the conversion effect in percentage points, the unpooled standard error, and the contribution calculation. Check the arithmetic independently. Ask it to explain why restricting the sample to email openers changes the analysis.

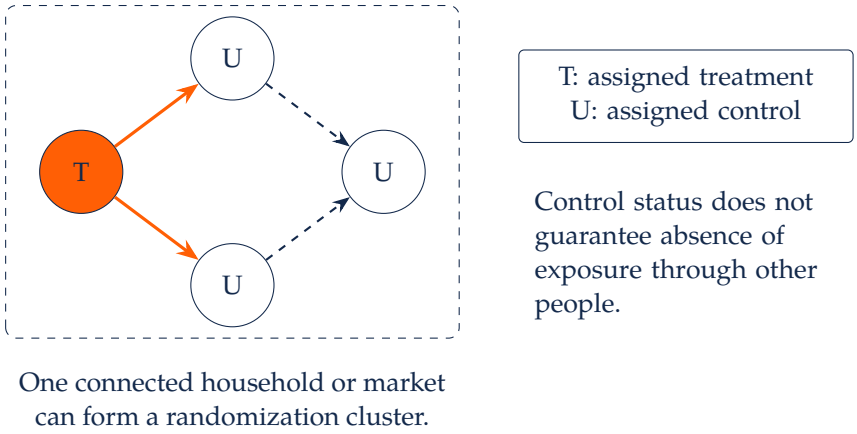


Figure 6.3. A treatment can affect both assigned units and nearby untreated units.

For observational data, draw or describe the assignment process before selecting controls. Request an overlap diagnostic and a report of which observations receive large weights. For DiD, ask the assistant to show observed and counterfactual series separately and identify the assumptions that connect them. A convincing plot should not substitute for those conditions.

If comparison-market sales rise by 25 rather than 10, the DiD estimate becomes five. If email cost doubles to $1, the point estimate of incremental contribution falls to $0.20. If treatment raises returns, explain why a conversion effect can overstate the contribution effect. If two stores share most customers, identify why store assignment may create interference.

6.12 Readings

Facure's causal-inference handbook supplies accessible implementations of several designs. Kohavi and colleagues' practical guide discusses controlled online experiments. Callaway and Sant'Anna (2021) develop difference-in-differences with multiple periods.

Chapter 7

Prediction, Machine Learning, and Marketing Actions

A retailer wants to anticipate next month's customer purchases and allocate limited service capacity. A model that predicts historical purchases accurately may still fail when deployed: its features may contain future information, its test set may reuse the same customers as training, or its objective may reward errors that are costly for the business. Prediction should therefore be designed around an explicit deployment situation.

7.1 Specify the Forecast Origin and Target

A prediction task needs a target variable, a forecast origin, a horizon, and an eligible population. For customer churn, define what counts as churn and when that status becomes observable. In a subscription business, cancellation may be explicit. In noncontractual retail, "no purchase for 90 days" is an operational label, not proof that a relationship has ended permanently.

Construct predictors using information available at the forecast origin. If predicting purchases during November, features might use transactions through October 31. November support contacts, returns, or campaign interactions are unavailable at that origin. A database may contain them when the analyst extracts the file, making leakage easy to overlook. Store feature timestamps or document their measurement windows.

The deployment population determines the validation design. Predicting future behavior for existing customers differs from predicting behavior for newly acquired customers. A random row split can place the same customer in training and test data, giving a favorable assessment for a task that actually requires generalization to new customers. A time split addresses future periods, while a customer-group split addresses new customers. Some tasks require both.

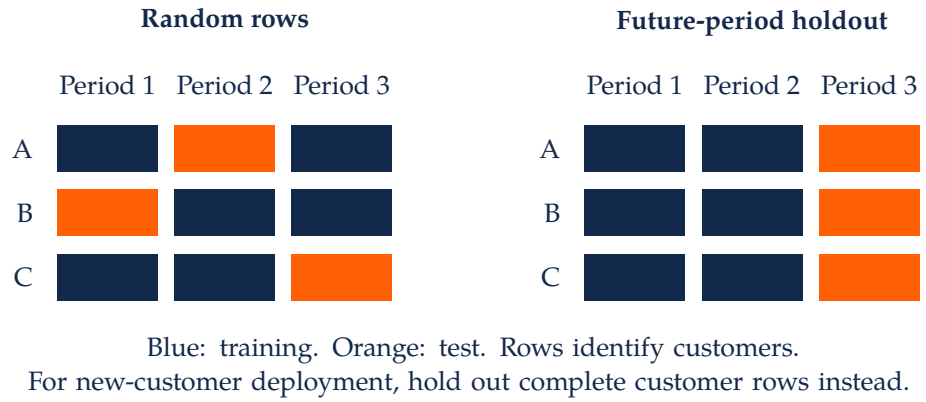


Figure 7.1. The validation split should reproduce the information available at deployment.

A random row split can reuse a customer's history in training and test data. Holding out complete customers evaluates generalization to new customers; holding out later periods evaluates future performance. The intended use determines which comparison is relevant.

7.2 Baselines, Loss Functions, and Regularization

A baseline should be operationally meaningful. For sales forecasting, compare with the previous period, the same period last year, or a simple customer-history average when those quantities are available. For classification, compare with a constant probability and a transparent logistic regression. A complex model's incremental value is easier to assess when these comparisons are reported on the same test observations.

A supervised learner minimizes a loss that quantifies prediction error. Squared error penalizes large deviations strongly; absolute error gives a different compromise under skewness. For a binary outcome, log loss is

$$L = -\frac{1}{n}\sum_{i=1}^{n}\{y_i \log \hat{p}_i + (1 - y_i)\log(1 - \hat{p}_i)\}.$$

Here y_i is the observed binary outcome and $\hat{p}_i$ is its predicted probability. Confident mistakes receive large penalties. The loss evaluates probability forecasts; it does not by itself encode the cost of a marketing action. Select a loss appropriate to the prediction task, then evaluate how the forecasts support the downstream decision.

Regularization constrains a flexible model to reduce overfitting. Ridge regression adds a squared-coefficient penalty, while lasso adds an absolute-coefficient penalty and can set some coefficients to zero. Standardize features appropriately when the penalty depends on coefficient scale, and fit that transformation using training data. The regularization strength is a tuning choice selected with validation data, not with the final test sample.

Bias and variance describe an important tradeoff. A rigid model may systematically miss a pattern; a very flexible model may fit noise specific to the training sample. More data can help, but only if it represents the deployment problem and adds relevant information. A million duplicated or highly dependent rows can provide less useful information than their count suggests.

7.3 Logistic Regression, Trees, and Ensembles

Logistic regression maps a linear index to a probability:

$$\hat{p}_i = \frac{1}{1 + \exp[-(\beta_0 + x_i^\top \beta)]}.$$

The coefficients describe changes in log odds under the model. A coefficient's effect on probability depends on the starting probability and other predictors. Predicted probabilities are often easier to communicate than odds ratios, especially when the business decision uses expected counts or contribution. Nonlinear features and interactions can extend the model while preserving a relatively transparent structure.

A decision tree partitions the feature space into regions using sequential splits. It can represent interactions without specifying them manually: a spending threshold might matter only among recent customers. Deep trees can fit small idiosyncratic groups, making pruning, minimum leaf sizes, and validation important. A split point near an observed value should not automatically be interpreted as a behavioral threshold.

Random forests average trees fitted with randomness in observations and candidate features, reducing instability relative to a single tree in many settings. Gradient boosting builds an ensemble sequentially to improve a chosen loss, with tuning choices such as tree depth, learning rate, and number of iterations. These methods can be effective on tabular data, but performance remains empirical. More complexity also increases the burden of checking leakage, calibration, computation, and reproducibility.

Neural networks learn layered representations and can be especially useful for text, images, or large complex datasets. Their flexibility does not eliminate the need for baselines or careful evaluation. In a modest customer table, a well-tuned simpler model may be competitive and easier to maintain. Choose the method based on task performance and practical constraints rather than the apparent modernity of the algorithm.

7.4 Validation, Cross-Validation, and Leakage

Use training data to fit parameters, validation data or cross-validation to select models and tuning choices, and a final test sample to estimate performance after those choices are fixed. Reusing the test sample repeatedly turns it into a validation sample, even if the file is still named "test." Report the procedure so the claimed holdout remains meaningful.

All learned preprocessing belongs inside the training process. This includes imputation, scaling, feature selection, vocabulary construction, dimensionality reduction, and target-based encodings. In cross-validation, fit each transformation on the training fold and apply it to that fold's validation observations. A pipeline makes the sequence explicit. The scikit-learn pitfalls guide documents implementation patterns for this separation.

Time-series and grouped data need adapted splits. Rolling-origin evaluation trains on earlier periods and predicts later ones, potentially repeating this across several origins. Grouped validation holds out complete customers, households, stores, or other relevant units. If a target window overlaps a training outcome window, a temporal gap may be necessary. Document the dates and units assigned to each split.

A useful audit asks when each feature became observable. In a marketing-call dataset, call duration can be useful after a call but unavailable when deciding whom to call. A model using that feature answers a different prediction problem. The feature's legitimacy follows the decision time, not whether a package accepts it or whether it improves accuracy.

7.5 Case Study: Accuracy Can Conceal a Useless Model

Suppose 5% of 1,000 customers make a purchase in the target window. A model predicting no purchase for everyone achieves 95% accuracy while identifying no purchasers. A different model

flags 100 customers, including 30 purchasers and 70 nonpurchasers. It has 30 true positives, 20 false negatives, 70 false positives, and 880 true negatives.

The second model's precision is 30%, recall is 60%, and accuracy is 91%. Its lower accuracy coexists with a potentially useful ranking for a limited-capacity task. Whether it improves profit depends on the action, costs, and whether the purchase probabilities under that action are the relevant quantities. Accuracy alone cannot make that decision.

ROC curves compare true-positive and false-positive rates across thresholds. Precision-recall curves focus on the tradeoff between finding positives and the composition of the flagged group, which can be informative when positives are rare. Both depend on the evaluation setting; precision is especially sensitive to prevalence. Lift at a specified contact fraction compares the selected group's outcome rate with the population rate, but observed lift is not automatically an incremental treatment effect.

Report metrics with their operational units. For a service team that can review only 100 cases per week, precision among the top 100 and the number of important cases missed may be more relevant than a global ranking metric. Use a holdout that reflects the weekly population and inspect uncertainty in the metric when counts are small.

7.6 Ranking and Calibration Answer Different Questions

A ranking model orders cases from more likely to less likely. A calibrated probability model assigns probabilities that correspond to observed frequencies in relevant groups. If customers assigned probability 0.20 purchase about 20% of the time in a suitable evaluation population, that group is calibrated. Calibration can vary across segments and over time even when overall performance looks satisfactory.

A strictly increasing transformation of scores preserves their ranking. Squaring probabilities between zero and one therefore leaves a ranking-based AUC unchanged, apart from practical handling of ties or finite precision, while generally changing calibration. A predicted probability of 0.50 becomes 0.25. If the original 0.50 group purchases half the time, the transformed probabilities understate its frequency.

For calibrated forecasts, predicted probabilities equal observed frequencies. After squaring an initially calibrated score p, the reported probability is $q = p^2$, while the observed frequency remains $p = \sqrt{q}$. Expected counts and profit calculations depend on the probability scale, so unchanged ranking performance does not guarantee unchanged decision quality.

Inspect calibration plots and proper scoring rules, such as log loss or the Brier score, on held-out data. Calibration methods themselves require fitting and should use data separate from the final evaluation. If the deployment population changes, recalibration may be necessary. A single global correction may be insufficient when the relationship changes differently across groups.

7.7 Thresholds Should Reflect the Decision

Consider a classification action in which a false positive costs C_{FP} and a false negative costs C_{FN}, with zero cost for correct classifications. For a calibrated probability p, predicting positive has

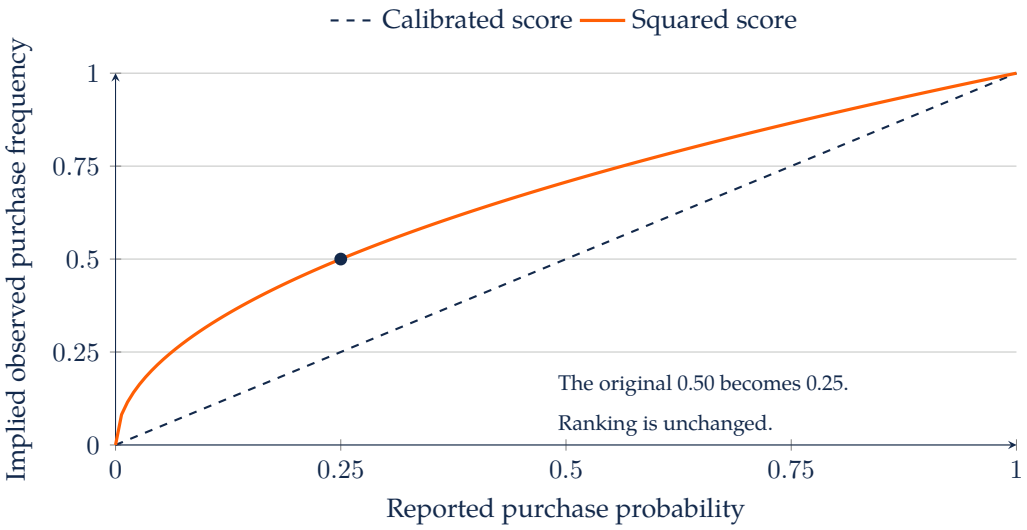


Figure 7.2. A monotonic score transformation preserves ranking while changing probability calibration.

expected cost $C_{FP}(1-p)$ and predicting negative has expected cost $C_{FN}p$. The cost-minimizing rule predicts positive when

$$p > \frac{C_{FP}}{C_{FP}+C_{FN}}.$$

If an unnecessary review costs one unit and missing a true issue costs nine units, the threshold is 0.10 under this simplified loss. A threshold of 0.50 would encode a different cost balance. Capacity constraints, review effectiveness, and heterogeneous costs require a more detailed rule. The formula is useful because it makes the decision assumptions explicit.

A marketing intervention usually needs a causal comparison beyond this classification loss. For a coupon, let $p_i(1)$ and $p_i(0)$ be purchase probabilities with and without the coupon, m the contribution before the discount, d the discount paid on a treated purchase, and c the contact cost. A simple incremental expected contribution is

$$\Delta\pi_i = m\{p_i(1)-p_i(0)\} - d\,p_i(1) - c.$$

The discount is paid on all treated purchases in this setup, including purchases that would have happened anyway. The equation assumes one relevant purchase opportunity and a fixed discount and contribution. It can be extended to spending, multiple purchases, returns, and heterogeneous margins.

7.8 Case Study: Targeting the Most Responsive Customers Can Lose Money

Take $m = 40$, $d = 5$, and $c = 2$. Customer A has purchase probabilities 0.80 with the coupon and 0.78 without it. Customer B has probabilities 0.30 and 0.15. A has the higher treated response probability, but incremental contribution is \$-5.20 for A and \$2.50 for B.

Customer	Purchase With Coupon	Purchase Without Coupon	Incremental Contribution
A	0.80	0.78	-$5.20
B	0.30	0.15	$2.50



Figure 7.3. Baseline purchasing and discount expense can reverse a response-based targeting rule.

Incremental purchasing contributes $0.80 for A and $6.00 for B, while discount and contact costs are $6.00 and $3.50, respectively. These values follow from the assumed purchase probabilities. Estimating their difference from customer data requires treatment variation and an identification strategy. Uplift or heterogeneous-treatment-effect models address that causal target under their conditions; they are not ordinary response models with a different name.

Evaluate a targeting policy on data that support its value comparison. A randomized holdout or a carefully designed policy evaluation can estimate incremental contribution for the selected group. Estimating a model and reporting its value on the same observations used to choose the policy can overstate performance. Budget constraints also matter: a customer with positive expected value may still be excluded when another customer offers greater value per unit of scarce capacity.

7.9 Interpretation, Fairness, and Deployment

Feature importance describes a relationship between a fitted model and its inputs under a chosen method. Permutation importance measures the change in predictive performance after disrupting a feature, with complications when predictors are correlated. Partial-dependence plots average predictions over a reference distribution and can involve unrealistic combinations of correlated features. Local explanation methods describe the fitted prediction under their conventions. These tools do not automatically identify causal effects of changing a feature.

Evaluate performance across groups relevant to the application and examine how errors translate into service or opportunity. A model can be accurate on average while performing poorly for new customers, small stores, or a language group with sparse training data. Protected or sensitive

attributes and proxies require appropriate institutional and legal review for the intended use. The consequences of errors depend on the decision and the affected customers.

Deployment requires monitoring the input distribution, missingness, outcome frequency, calibration, and decision value. A change in model accuracy can reflect changing customer behavior, instrumentation, or policy. Record versions and retain a reproducible baseline. If an intervention changes the behavior that generates future training data, the model's environment becomes endogenous to its use, because its recommendations change subsequent exposure and behavior.

7.10 AI-Assisted Workflow, Practice, and Readings

Ask an assistant to create an evaluation plan before choosing a model. Specify forecast origin, horizon, unit, grouping, baseline, loss, and final holdout. Require a feature-availability table and a pipeline that fits transformations only on training data. Have it produce both a metric table and an explanation of the operational decision each metric supports.

For practice, verify the 91% accuracy and 60% recall in the rare-purchase example. Square a small set of predicted probabilities and confirm that their order is unchanged while expected purchase counts change. Recalculate the coupon case with a $1 discount and explain which customer becomes more attractive. Finally, identify why a model trained on all customers' latest outcomes cannot honestly claim to predict those same outcomes out of sample.

An Introduction to Statistical Learning develops the central learning methods. The scikit-learn evaluation guide provides metric definitions and implementation references. Hagen and colleagues (2020) connect machine-learning tools to behavioral marketing questions.

Chapter 8

Current Topics and Integrated Marketing Decisions

A marketing team can now generate campaign copy, build a prediction model, and summarize customer feedback within a short working session. The remaining difficulty is deciding whether the resulting system improves a business decision. A recommendation engine can reproduce past exposure, a media model can attribute seasonal demand to advertising, and an AI-generated analysis can calculate an invalid comparison precisely. Evaluating these systems requires a clear account of their information, incentives, assumptions, and consequences.

8.1 Recommendation Systems Learn From Selected Exposure

A retailer wants to recommend products that customers will find useful. A common starting point is past interaction: purchases, clicks, ratings, or time spent. These observations combine preferences with the opportunities customers received. A product displayed prominently has more chances to be clicked than an otherwise similar product that was rarely shown. An absent interaction can therefore mean dislike, lack of exposure, lack of need, or an unavailable product.

Content-based methods use product attributes and customer histories to identify similar items. Collaborative methods use patterns across users and items, including matrix factorization, which approximates an interaction matrix with lower-dimensional user and item representations. Hybrid systems combine these sources. Each method needs a strategy for new customers and new products that have little interaction history. Popularity baselines and simple category rules provide useful comparisons.

Recommendation systems also change the data used to train the next model. A product that receives more exposure can accumulate more interactions, become easier to estimate, and receive still more exposure. This process can amplify initial differences without establishing that the repeatedly displayed product would be preferred under equal exposure.

Rankings determine exposure, exposure affects observed behavior, and that behavior enters subsequent training data. Customer preferences also affect behavior, while availability and position affect the opportunity to respond. Logging impressions, positions, eligible alternatives, and the policy that selected them helps distinguish these components. Purchase logs alone often omit the alternatives a customer could actually have encountered.

Evaluate recommendation quality at the unit relevant to the decision. Precision among the top recommended items and ranking metrics assess aspects of relevance. Coverage, novelty, diversity, repeat engagement, returns, and contribution describe other consequences. A system can improve one metric while reducing another. Offline evaluation on historical logs is particularly sensitive to the policy that generated those logs. An experiment that changes exposure can supply information that a more flexible model of the same selected data cannot supply.

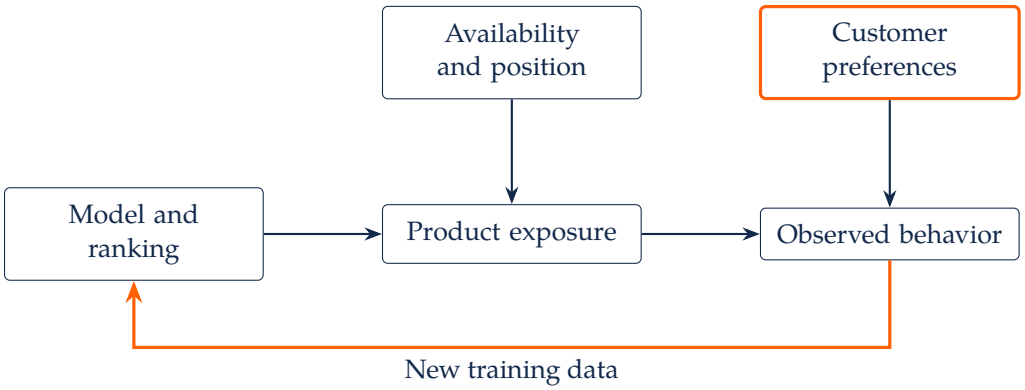


Figure 8.1. A recommender changes the exposure that generates its next training sample.

8.2 Case Study: Relevance, Variety, and Customer Goals

Nielsen, Yang, Dubé, Knäuper, Ling, and Nie study consumption variety in food recommendation. Their research connects observed patterns in food logs with a recommendation method that incorporates variety. Ling and coauthors develop a goal-oriented recommendation approach using behavior from comparable users. The choice of objective matters: reproducing observed choices can conflict with a customer's stated goal. Observational relationships and offline ranking evaluations measure different objects from the causal effect of deploying a recommendation policy.

Suppose a retail customer requests three products and values variety across categories. Assign four candidate items the relevance scores below. The scores are comparable across these items and measure relevance rather than purchase probability.

Item	Category	Relevance Score
A	Coffee	0.90
B	Coffee	0.85
C	Tea	0.75
D	Fruit	0.65

Choosing the three highest scores gives A, B, and C, with total relevance 2.50 and two categories. Choosing A, C, and D gives relevance 2.30 and three categories. To make the tradeoff explicit, define a list objective

$$J(L) = \sum_{j \in L} s_j + \lambda K(L),$$

where L is a three-item list, s_j is relevance, $K(L)$ counts distinct categories, and λ is the value placed on an additional category in the same units as the relevance score. With $\lambda = 0.30$, A, B, C has objective 3.10, while A, C, D has objective 3.20. The latter list is preferred under this stated objective. The threshold at which these two lists tie is $\lambda = 0.20$.

The variety weight must come from an explicit objective, user preference, or evaluation process. Category counts are also a coarse measure: they treat all category differences equally and ignore which products the customer already owns. A deployed system would need better measurements and an evaluation of whether the resulting lists improve the intended customer experience.

For an experiment, randomize eligible customers to the existing and revised recommendation policies. Choose a primary outcome such as contribution per assigned customer over a fixed horizon, alongside measures of customer satisfaction and unwanted repetition. Analyze assignment rather than only customers who clicked a recommendation. Clicking occurs after the policy is assigned and can differ between policies. Preserve the list shown and the alternatives eligible at each impression so the mechanism can be examined.

8.3 Marketing-Mix Models Combine Carryover and Saturation

A marketing-mix model relates aggregate outcomes to media activity and other determinants across time or markets. Its appeal is practical: many firms observe weekly sales and channel spending even when individual customer paths are unavailable. The analytical challenge is that media decisions often respond to expected demand. Advertising, promotions, holidays, distribution, and competitor actions can move together.

A simple carryover variable is

$$a_{ct} = s_{ct} + \rho_c a_{c,t-1}, \qquad 0 \leq \rho_c < 1,$$

where s_{ct} is spending in channel c during period t, a_{ct} is its accumulated exposure proxy, and ρ_c controls geometric decay. This unnormalized definition gives the same units as spending, but its long-run level under constant spending is $s_c/(1-\rho_c)$. Some software normalizes the weights. Coefficients and interpretations must follow the selected convention.

If one spending pulse of 100 occurs at period zero and $\rho = 0.5$, the subsequent accumulated values are 100, 50, 25, and 12.5. A pulse can therefore coincide with later sales even when current spending is zero. A seasonal demand pattern can also create persistence. The model needs enough information and a credible design to distinguish these explanations.

Saturation describes declining additional response at higher exposure. A saturating response curve is $R(a) = A[1-\exp(-a/b)]$, where A is the limiting response and $b > 0$ controls how quickly the curve approaches that limit. A basic outcome model may combine channel response curves, controls, trend, seasonality, and an error term. The fitted components are model-dependent; their causal interpretation requires assumptions about how spending was chosen and which demand determinants were measured.

The official Meridian documentation describes a modern framework for marketing-mix modeling. The PyMC Marketing examples provide another implementation route. Use those materials to inspect definitions, priors, and diagnostics. A software package cannot supply a missing source of identifying variation. Experiments and external evidence can help calibrate a model, subject to whether their populations, interventions, and time horizons match the intended use.

8.4 Case Study: Allocate the Next Dollar Using Marginal Returns

Consider two channels with imposed contribution response curves

$$R_A(s_A) = 100[1 - \exp(-s_A/20)], \qquad R_B(s_B) = 80[1 - \exp(-s_B/40)].$$

Both spending and contribution are measured in thousands of dollars over the same horizon. These response functions omit cross-channel interactions and are assumed known. At a total budget of 60, suppose the current allocation is 40 to A and 20 to B. The question is whether moving a small amount between channels increases contribution while holding the total budget fixed.

Average contribution per unit spent is $R_c(s_c)/s_c$. At the current allocation, it is approximately 2.16 for A and 1.57 for B. Marginal contribution is the derivative of the response curve. It is approximately 0.68 for A and 1.21 for B. A has the higher average return, while B offers the greater return on the next small unit of spending.

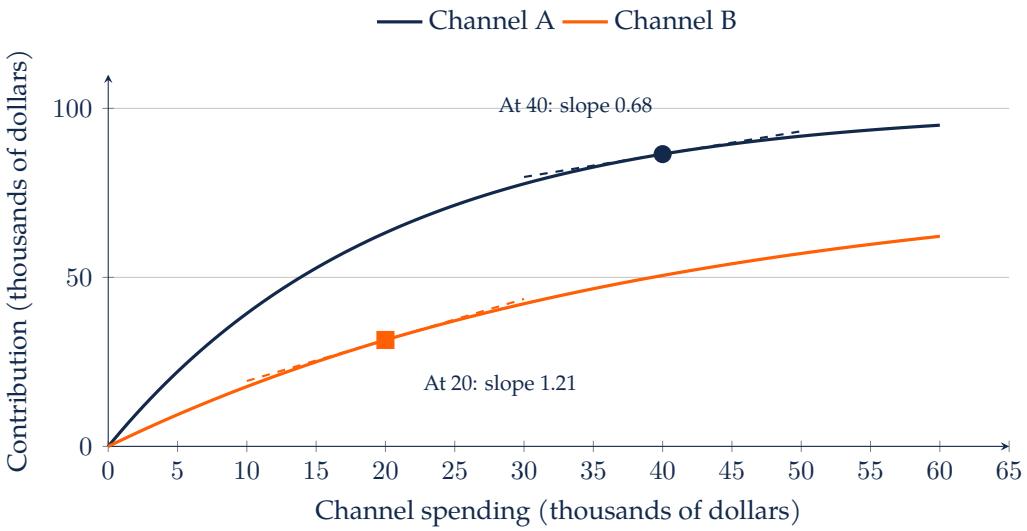


Figure 8.2. Average returns can favor one channel while marginal returns favor the other.

At the current allocations, B has the steeper response curve. Moving a small amount from A to B therefore raises total contribution. The comparison depends on the slopes at the current allocation, whereas average returns summarize all spending up to that point. The budget is held fixed, so the transfer does not change total spending expense.

For a differentiable, concave response with an interior optimum and no other constraints, marginal returns are equal across channels. Here the condition is

$$5\exp(-s_A/20) = 2\exp(-s_B/40), \qquad s_A + s_B = 60.$$

Solving gives $s_A \approx 32.22$ and $s_B \approx 27.78$. Total contribution rises from approximately 117.94 to 120.09, or about 2.14 thousand dollars. If total budget were also a choice, marginal contribution would be compared with the marginal cost of spending; exhausting a fixed budget and choosing its size are different optimization problems.

A real decision should also examine parameter uncertainty, operational minimums, saturation outside observed spending ranges, delayed effects, and interactions across channels. Report how the recommended allocation changes under plausible response curves. A sharply stated optimum based on weakly identified slopes can conceal substantial decision uncertainty. Prefer an allocation that is supported by observed variation and specify an experiment that would resolve the most consequential uncertainty.

8.5 Generative AI Changes the Cost of Producing and Checking Work

Generative AI can assist with code, text classification, documentation, and communication. The business value depends on the complete workflow. Faster generation can be offset by additional review, integration failures, or errors that reach downstream decisions. Measure the output that matters: a correct, reproducible analysis completed with an appropriate understanding of its limitations.

Separate several uses. A coding assistant produces or edits a program that can be inspected and executed. An annotation model produces labels that require a measurement evaluation. A retrieval system selects source passages and may generate a summary that needs factual support. A content system produces customer-facing material whose effect should be evaluated in its deployment context. Each use has a different unit of quality and a different failure mechanism.

For coding, begin with an analytical specification and a small example with a known result. Ask the assistant to implement a limited step, explain the data transformations, and identify assumptions. Inspect the code before running it. Then compare intermediate outputs with independent checks. A program that executes successfully can still use an incorrect denominator, join on the wrong key, or include future information. Those errors require substantive checks rather than syntax repair.

For retrieval and summarization, maintain the connection between a claim and its source. A citation should support the particular statement next to it. Ask the model to distinguish an observed result from a proposed explanation, and check that distinction against the original text. If a passage or file is unavailable, state that limitation. Fluent reconstruction is not a substitute for access to the evidence.

8.6 Case Study: Evaluate an AI-Assisted Analytics Workflow

Suppose a firm wants to know whether access to a coding assistant improves analysts' work on customer-retention tasks. Define treatment as access under a specified protocol and version, and define an outcome that incorporates both completion and validity. A useful primary outcome could be the fraction of assigned tasks meeting a prespecified analytical rubric within a fixed time budget. Secondary outcomes could include completion time, independent error severity, and a later explanation task completed without assistance.

Randomize analysts or teams according to the likely spillovers. Repeated tasks by the same analyst require attention to learning, carryover, and dependence. If a crossover design is used, randomize order and consider whether experience in the first condition changes performance in

the second. A task-level comparison with no accounting for analyst ability or task difficulty can confound the treatment effect with who received which work.

Quality review should use a common rubric and, where feasible, reviewers unaware of treatment assignment. The rubric can check observation units, joins, leakage, estimation, uncertainty, and interpretation. Record the assigned task, the eligible inputs, the tool configuration, and the evaluation procedure. Avoid interpreting a favorable result for one task type as a universal effect across all analytical work.

Suppose a manual workflow uses 90 minutes to produce an analysis and 20 minutes to review it, totaling 110 minutes. An assisted workflow uses 45 minutes to produce a draft and 40 minutes to review it, totaling 85 minutes. The reduction in total time is 25 minutes, or about 22.7%, while the drafting-time reduction is 50%. The total-time comparison is informative only if the completed outputs meet comparable quality requirements.

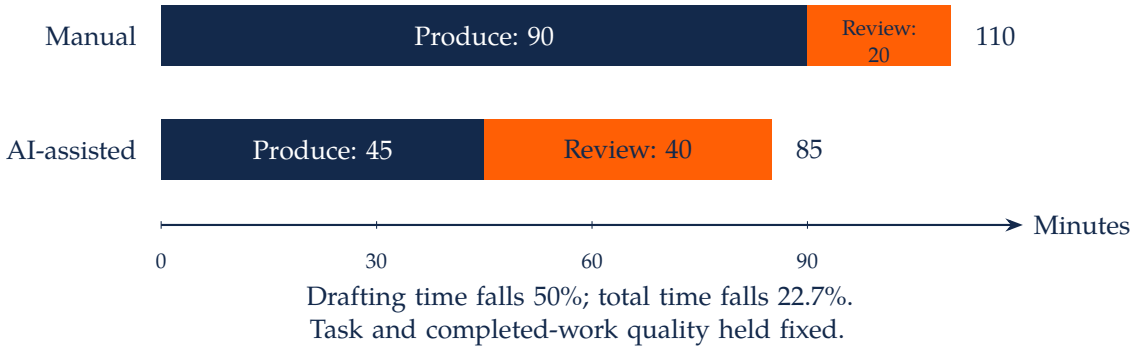


Figure 8.3. The relevant time saving includes generation, review, and correction.

This comparison holds the task and required quality fixed. Review is part of production because an undetected error can alter a marketing decision. In empirical work, report the completion and quality outcomes alongside time so that faster abandonment or lower-quality work is not counted as an equivalent improvement.

8.7 Integrated Case: A Retail Subscription Launch

A retailer is considering a subscription service with recommendations and delivery benefits. Begin with the decision: which offer to launch, to which eligible customers, at what price, with what evidence of incremental contribution? Define the evaluation horizon, capacity constraints, and the alternatives to launch.

Construct the customer and order tables with explicit keys and dates. Describe customer activity and purchasing cohorts, keeping unavailable future retention separate from observed inactivity. Use regression to summarize conditional patterns while examining confounding. Design a conjoint exercise to learn stated tradeoffs between delivery, assortment, and price, then assess whether its predictions generalize to an actual offer.

Use segmentation to organize meaningful differences in needs and behavior, and calculate customer value from contribution rather than revenue. Analyze reviews to identify the service attributes customers mention, checking annotation quality against human labels. Develop a pre-

diction model using only information available at the targeting date. Its forecasts can help plan capacity, while a randomized offer evaluates incremental take-up, contribution, and retention.

Finally, evaluate the recommendation component and the acquisition budget. Log exposure to recommended items, distinguish average from marginal media returns, and use AI assistance within a documented verification process. The management recommendation should identify the supported action, the uncertainty that could change it, and the next observation or experiment with the greatest decision value. A complicated collection of models is useful only if these pieces support a coherent choice.

8.8 Practice and Further Reading

Recommendation exercise. Compare all four three-item lists in the variety example at $\lambda = 0$, 0.20, and 0.30. At zero, A, B, C is optimal. At 0.20, it ties A, C, D. At 0.30, A, C, D is optimal. Explain how a different category definition could change that conclusion without changing any relevance score.

Budget exercise. Calculate the gain from moving one thousand dollars from channel A to channel B at the original allocation. Compare the exact change with the difference between the two marginal returns. The derivative is a local approximation; the exact finite transfer also reflects curvature. Then explain why an unobserved seasonal demand shock could invalidate a causal interpretation of an estimated response curve.

AI evaluation exercise. Draft a rubric for an assistant-generated retention analysis. Include at least one check that detects a substantive error in code that executes successfully, such as an order-line denominator used for average order value. Define a hand-checkable example that exposes the error and explain how a reviewer would determine whether it occurred.

Read Nielsen and coauthors, "Consumption Variety in Food Recommendation", and Ling and coauthors, "Following Good Examples", for applications connecting recommendation objectives with customer behavior. The official marketing-mix documentation linked above develops implementation details. Consult the AI-coding evidence and workflow resources for further study.